A Maintenance-Driven Reduction Gearbox for a Rear-Drive Pusher Turboprop

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This paper presents the complete design of the power transmission of a 1,200 shp (895 kW) pusher turboprop derived from a T700-class turboshaft core, carried from requirements to a verification plan in four review-gated phases and explained so that a reader new to gearboxes can follow every decision. The customer case is a medium-altitude long-endurance unmanned aircraft of the Heron TP class that flies today on a PT6A-67A. A rear-drive installation keeps the natural gas path of the straight-through core; an offset helical first stage (31/110 teeth) followed by a four-planet planetary stage (45/37/119 teeth, fixed ring) gives the 12.93:1 reduction from 22,000 rpm to 1,700 rpm without a fast planetary stage, so that rolling planet bearings and an emergency oil-mist run-dry capability become possible. Gears were rated to ISO 6336:2019 (lowest pitting safety 1.28, root safety 1.55 with dynamic loads), bearings to ISO 281 (lowest life 14,500 h against 12,000 h), the cases and carrier by three-dimensional finite elements, and the forced response of the gear webs and cases to the transmission error was closed with a friction damping ring. A fault tree keeps every propeller-release cut set below 10^{-8} per flight hour, a 100-mode failure mode and effects analysis identifies seven critical parts, and damage-tolerance lives exceed their targets. Materials, tolerances (ASME Y14.5) and representative drawings define manufacture; a synthetic-signal study sets condition-monitoring indicators and thresholds; and 25 planned tests close the 69 requirements of the final baseline. The gearbox weighs 109.95 kg, which places it at the top of the CT7-9 trend scaled to the same torque. Open issues are stated: a near-zero mass margin, an under-estimated mount reaction in one interface value, unproven fire sealing, and a CAD model that lags the latest analysed geometry.

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Abbreviations

AGB	=	accessory gearbox
APR	=	automatic power reserve
BOM	=	bill of materials
CAD	=	computer-aided design
CDR	=	critical design review
CS-E	=	EASA Certification Specifications for Engines
DT	=	damage tolerance
EASA	=	European Union Aviation Safety Agency
ENV	=	envelope (condition indicator)
FADEC	=	full authority digital engine control
FE	=	finite element
FMEA	=	failure mode and effects analysis
FTA	=	fault tree analysis
GD&T	=	geometric dimensioning and tolerancing
HIP	=	hot isostatic pressing
HUMS	=	health and usage monitoring system
ICA	=	instructions for continued airworthiness
ICD	=	interface control document
LTCA	=	loaded tooth contact analysis
MALE	=	medium altitude, long endurance
MCP, MCT	=	maximum continuous power, torque
NDT	=	non-destructive testing
ODM	=	oil debris monitor
OSG	=	overspeed governor
PCU	=	propeller control unit
PDR	=	preliminary design review
PT	=	power turbine
QJ	=	four-point contact ball bearing
RGB	=	reduction gearbox
SER	=	sideband energy ratio
SFC	=	specific fuel consumption
SRR	=	system requirements review
TBD	=	to be determined
TBO	=	time between overhauls
TBR	=	to be reviewed
TCDS	=	type certificate data sheet
TE	=	transmission error
TSA	=	time-synchronous average
UAV	=	unmanned aerial vehicle

Nomenclature

a	=	centre distance, mm
b	=	face width, mm
C, C_0	=	basic dynamic and static load rating of a bearing, N
d	=	pitch diameter, mm
f	=	frequency, Hz
f_{ZCa}	=	flank-modification factor of the contact-stress rating, –
FM4, NA4	=	kurtosis-based gear condition indicators, –
I_p, I_d	=	polar and diametral mass moment of inertia, $kg \cdot m^2$
i	=	overall reduction ratio, –
K_A, K_v	=	application and dynamic factors of ISO 6336, –

$K_{H\beta}$ = face load factor, –
K_{γ} = planet load-sharing factor, –
L_{10m} = modified rating life of a bearing, h
m = module, mm
N = number of planets, –
n = rotational speed, rpm
P = power, kW
P_{fa} = false-alarm probability per record, –
S_F, S_H, S_S = safety factors against tooth-root breakage, pitting and scuffing, –
T = torque, N·m
z = number of teeth (subscripts s sun, p planet, r ring, 1 pinion, 2 gear)
β = helix angle, deg
ζ = damping ratio, –
λ = failure rate, h^{-1}
σ = stress, MPa
τ = torque decay time constant, s
Ω, ω = rotational speed, rad/s

I. Introduction

A turboprop engine has two very different machines in it: a gas turbine that is happiest spinning at tens of thousands of revolutions per minute, and a propeller that must turn slowly so that its blade tips stay below the speed of sound. Everything between the two, the shafts, the reduction gearbox (RGB), its bearings and lubrication and the drives for the accessories, is the power transmission. This report describes the complete design of such a transmission, from the first list of requirements to a plan of the tests that would prove it.

The design case is a medium-altitude, long-endurance (MALE) unmanned aircraft of the Heron TP class [1]. It has one pusher propeller at the rear of the fuselage, flies missions of more than 30 hours and today uses a PT6A-67A turboprop of 1,200 shp (895 kW) [2]. The question the project asks is: if the engine were instead derived from a T700-class turboshaft core, the gas generator behind the CT7 turboprop family [3, 4], what power transmission would it need, and how good could it be? The core itself is treated as given; the work starts at the output of its power turbine and ends at the propeller flange (Fig. 1).

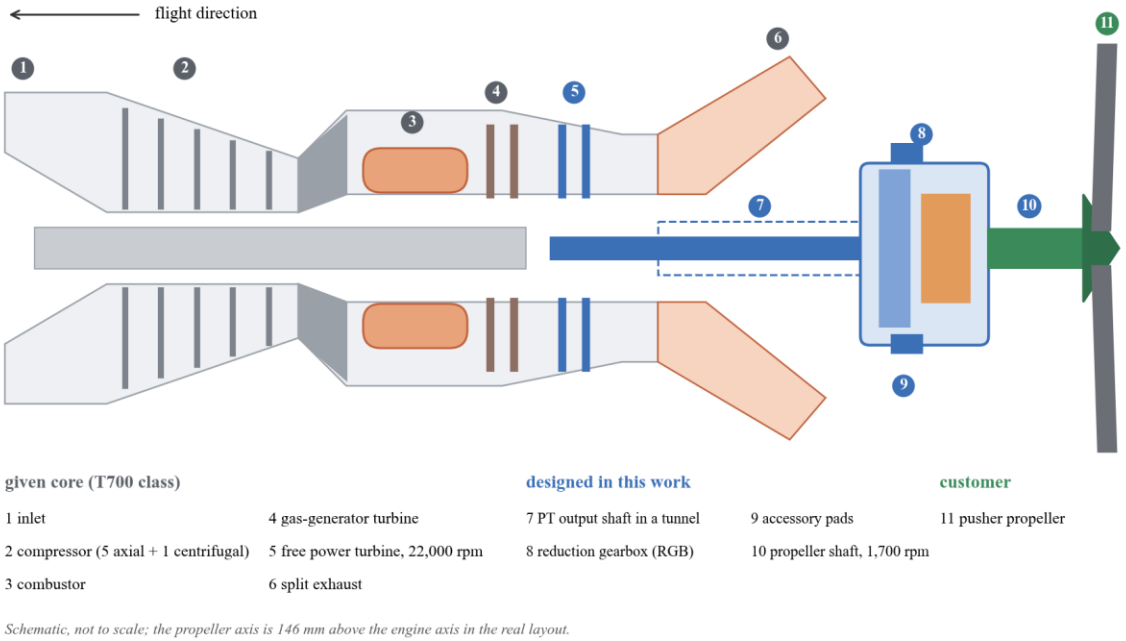


Fig. 1 The engine considered in this work. The grey core is taken as given; the blue parts, from the power-turbine output shaft to the propeller shaft, are designed here. What to look at: the power leaves the core at the rear, runs through a tunnel between the two exhaust branches and reaches the gearbox, which drives the pusher propeller.

Three facts make the problem interesting. First, the T700 is a straight-through engine whose power shaft leaves at the front, through the middle of the air inlet, while the propeller of a pusher aircraft is at the back. Second, the speed ratio needed, 22,000 rpm down to 1,700 rpm or 12.94:1, is lower than in any reference turboprop, which changes the usual choice of gear arrangement. Third, the customer is military and unmanned, so the gearbox must keep running for a while after losing its oil, must be quick to change in the field and should warn of damage before it becomes dangerous.

The work was organised like a small engine programme, in four phases that each ended with a review (Fig. 2): Phase 0 collected the literature and wrote the requirements (system requirements review, SRR); Phase 1 compared concepts and chose one (preliminary design review, PDR); Phase 2 designed and analysed every part (critical design review, CDR); and Phase 3 produced the manufacturing definition, the safety analyses and the verification plan. Every calculation was written as a MATLAB tool and checked against a separate Python implementation, the parts were modelled in CATIA V5 by scripts fed directly from the calculations, and every decision with a real trade-off was taken by the author and recorded (Appendix A).

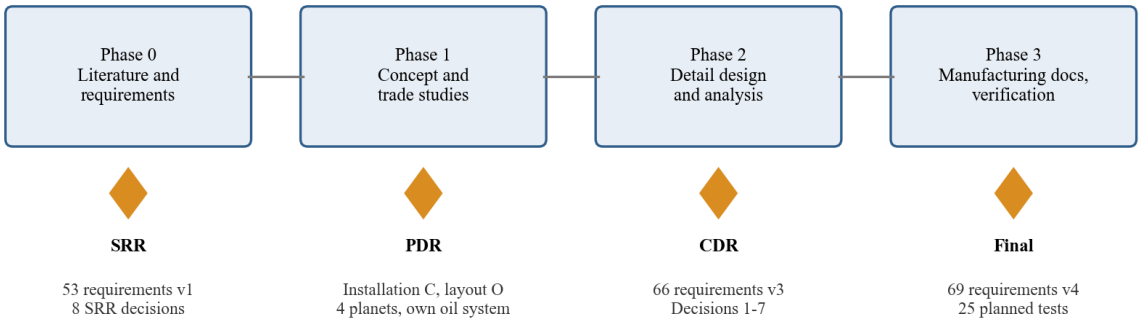


Fig. 2 The four phases, the reviews that closed them and their main outcomes.

The study is theoretical: no hardware is built, and manufacture is judged on paper against standard processes. Every calculation followed the same order. The method was written down first (inputs, assumptions, formulas); then test cases with known answers were prepared; only then was the tool coded. For the preliminary sizing, for example, nine test cases with 109 expected values came from the Python reference and every function had to reproduce them within 0.1%; MATLAB R2023b, GNU Octave 8.4 and Python then gave identical results. All inputs of a tool sit in one file, so that changing an assumption is a one-line edit and every number can be traced. Every change entered the project repository as a reviewed pull request (Appendix B).

Limits of the cross-check

Agreement between MATLAB, Octave and Python catches coding mistakes, such as a wrong index or a forgotten unit conversion, but not a wrong formula or coefficient, because the same error would enter all three. Such errors were found instead by comparing every coefficient clause by clause with the standards; in the gear rating this gave five corrections (Section IV.C). A fully independent check, through the solved examples of ISO/TR 6336-30 and a hand calculation of one case, is still recommended before the results are used further.

How to read this report

Each section starts with the question it answers and the answer in one or two sentences. Terms are explained where they first appear; the most important ones are also collected in Table 1. Shaded boxes like this one hold a key idea, a term or a short worked example; they can be skipped by a reader who knows the subject. Numbers in square brackets refer to the references at the end.

II. Basic Concepts

This section explains, without equations beyond the simplest ones, why a turboprop needs a reduction gearbox, why the torque grows as the speed falls, and how the planetary gear stage used in this design works. A reader familiar with gas turbines and gears can go straight to Section III.

A. How a Turboprop Delivers Power

The core of the engine is the gas generator: a compressor squeezes the air, fuel burns in the combustor, and a turbine takes just enough energy from the hot gas to drive the compressor. The gas that leaves it is still hot and under pressure, and it drives a second, **free power turbine** (PT). "Free" means that this turbine has no mechanical link to the gas generator; it only drives the load, here the propeller. In the T700 family the power turbine turns at 22,000 rpm at full power and delivers its power through a shaft that runs inside the gas generator [4, 5].

B. Why the Speed Must Be Reduced

A propeller loses efficiency and becomes very noisy when its blade tips approach the speed of sound, because shock waves form on the blades. The propeller of this aircraft has a diameter of 2.67 m. Fig. 3 shows how its tip speed grows with rotational speed: at 1,700 rpm the tip moves at about 238 m/s, 70% of the speed of sound at sea level, which is a normal value for a turboprop. At the power turbine's 22,000 rpm the tip would have to move at more than 3,000 m/s, which is physically impossible. The turbine, on the other hand, needs its high speed to produce its power from a small diameter. The reduction gearbox reconciles the two.

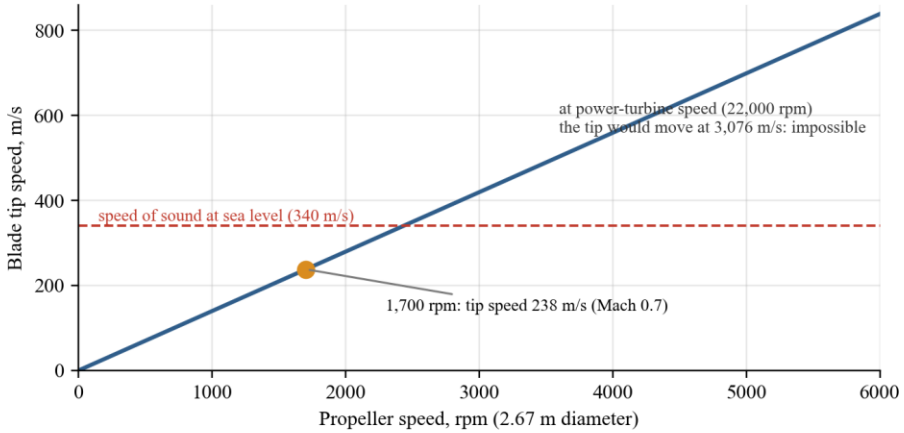


Fig. 3 Propeller tip speed against rotational speed for the 2.67 m propeller. The design speed of 1,700 rpm keeps the tips at Mach 0.7.

Worked example: tip speed

Tip speed $v = \pi D n / 60$. With $D = 2.67$ m and $n = 1{,}700$ rpm: $v = \pi \times 2.67 \times 1{,}700 / 60 = 237.7$ m/s. The speed of sound at sea level (15 °C) is 340 m/s, so the tip Mach number is 0.70. At $n = 22{,}000$ rpm the same formula gives 3,076 m/s, nine times the speed of sound.

C. Power, Torque and Speed

The power carried by a shaft is the product of its torque and its angular speed. If the same power is to be delivered at a lower speed, the torque must grow in proportion:

$$P = T\,\omega = T\,(2\pi\, n / 60) \quad \Rightarrow \quad T = 60\,P / (2\pi\, n) \tag{1}$$

With $P = 895$ kW the power-turbine shaft carries only 388.5 N·m at 22,000 rpm, but the propeller shaft carries 5,027 N·m at 1,700 rpm, about thirteen times more (Fig. 4). Since the size of a gear grows with the torque it carries, the last, slowest stage of the gearbox largely sets its size and mass. This simple fact lies behind many decisions in this project.

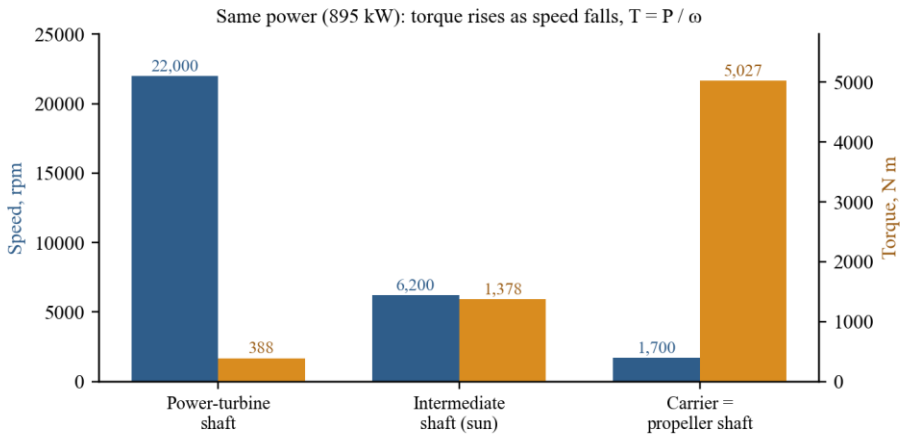


Fig. 4 Speed and torque of the three main shafts at full power. The power is the same on every shaft (losses neglected); the torque grows as the speed falls.

D. How a Planetary Gear Stage Works

A planetary (epicyclic) stage has a central **sun** gear, several **planet** gears that mesh with the sun and with an internal-toothed **ring** gear around them, and a **carrier** that holds the planet axles. In this design the ring is fixed to the case, the sun is the input and the carrier is the output: as the sun turns, the planets roll inside the ring and carry the carrier around with them, much more slowly than the sun. Because the load is shared by several planets, a planetary stage carries a large torque in a small diameter, and its input and output are on the same axis. Its ratio with a fixed ring is

$$i_{planetary} = 1 + z_r / z_s \tag{2}$$

where z_r and z_s are the numbers of teeth of the ring and the sun. The planets' own tooth number does not enter the ratio, but it decides whether they fit between the sun and the ring. An external gear pair (two gears meshing side by side) has the ratio z_2/z_1 and reverses the direction of rotation; it also moves the output to a parallel axis, which is called an **offset**.

> **Key idea: why the gearbox has two stages**
>
> 12.94:1 is too much for one gear pair: the small gear would be tiny and the large one huge. Two stages of about 3.6:1 each (since $3.6 \times 3.6 \approx 13$) keep every gear at a reasonable size. In the chosen design the first stage is an offset helical pair (110/31 = 3.55) and the second is a planetary stage (1 + 119/45 = 3.64).

E. Glossary of Key Terms

Table 1 Key terms used throughout the report.

Term	Meaning in plain words	Where it matters here
Reduction ratio	input speed divided by output speed	22,000 / 1,700 = 12.94
Efficiency	share of the input power that leaves the gearbox; the rest becomes heat in the oil	98.71% (12.4 kW of heat)
Safety factor	strength of the material divided by the actual stress; above 1 means margin	teeth: ≥ 1.2 against pitting, ≥ 1.5 at the root
Bearing life (L_{10m})	running time that 90% of identical bearings reach before fatigue damage	≥ 12,000 h, three overhaul intervals
Critical speed	a speed at which a shaft's rotation excites one of its own natural frequencies	kept ≥ 20% from the operating range
Natural frequency, mode	the frequency and shape in which a part vibrates by itself	kept away from tooth-meshing frequencies
Mesh frequency	how many tooth contacts happen per second: tooth number × rotation frequency	planetary 3,374 Hz, first stage 11,367 Hz
Finite element (FE) analysis	computing stresses and vibrations by dividing a part into many small elements	cases, carrier, planets, gear web
Critical part	a part whose failure would be hazardous; it gets an approved life and special control	seven parts, including the propeller shaft
TBO	time between overhauls	4,000 h
TBR, TBD	requirement value to be reviewed (provisional) or to be determined (unknown)	status of 41 of the 69 requirements

III. The Customer Case and Requirements

Before any part can be sized, the numbers it must meet have to be fixed and justified. This section explains where the design numbers come from: the customer aircraft, the core, the reference engines and the certification rules.

A. Scenario: Which Aircraft, Which Core

The customer is treated as an aircraft maker that hands the engine maker a set of needs, as happens in real programmes. Heron TP-class data are only public at brochure level: a maximum take-off mass of 5,670 kg, more than 30 hours of endurance, a ceiling above 45,000 ft, 220 kt maximum speed and a single 1,200 hp PT6 turboprop as a pusher [1]. The engine side is better documented, because the limits of certified engines are published in their type certificate data sheets (TCDS). Values not found in sources, such as the mission profile or the propeller inertia, were written down as explicit assumptions.

Designing a gas generator is a separate, very large task (compressor and turbine aerodynamics, combustion, cooling), so the T700-class core is taken as given and the whole effort goes into the mechanical transmission. The T700-GE-700 can give about 1,622 shp, 35% more than needed [4]. The engine is therefore **flat-rated**: the gearbox is designed for the torque of 1,200 shp, and the engine control holds the torque at this ceiling. The spare core capacity keeps the full power available on hot days and at altitude up to 53 °C, the same flat-rating temperature as the PT6A-67A. Table 2 collects the scenario numbers and where each comes from.

Table 2 The scenario numbers, their sources and why they were chosen.

Quantity	Value	Source	Why this value
Power	895 kW (1,200 shp)	PT6A-67A TCDS	power of today's engine; the aircraft is designed for it
Power-turbine speed	22,000 rpm	CT7-9 TCDS	PT speed of the T700 core in its turboprop version
Propeller speed	1,700 rpm	PT6A-67A TCDS	same propeller speed as today; subsonic tips
Reduction ratio	12.94:1 ± 0.5%	derived	integer tooth numbers cannot hit the ratio exactly
Propeller torque	5,027 N·m	derived	matches the 5,028 N·m of the PT6A-67A TCDS
Transient torque	6,915 N·m for 20 s	PT6A-67A TCDS	the aircraft and propeller are designed for it
Flat-rating temperature	53 °C	PT6A-67A TCDS	same hot-day performance as today

B. Reference Engines and Gearbox Arrangements

Table 3 compares the reference turboprops. All slow their power turbine in two stages, but in different ways: the PT6A-67A with two coaxial planetary stages [2], the PW100 family with an offset parallel-axis train, the CT7-9 with an integral propeller gearbox ahead of the core [3]. Seven arrangements were collected from patents and NASA studies: coaxial two-stage planetary, offset parallel axis, an offset first stage followed by a planetary stage [6], a tower-shaft offset drive [7], split torque [8], four layshafts [9] and a differential planetary for counter-rotation [10]. Our ratio of 12.94 is the lowest of all, because our power turbine is relatively slow and our propeller relatively fast.

Table 3 Reference turboprops and their reductions (ratios derived from the data-sheet speeds).

Engine	Layout	Input → propeller	Ratio
PT6A-67A	reverse flow; 2-stage planetary, coaxial	29,894 → 1,700 rpm	17.6:1
CT7-9B/C	front drive; integral gearbox ahead of the core	22,000 → 1,384 rpm	15.9:1
PW127	offset, double helical + spur	not given → 1,200 rpm	15.4–17.2:1
TPE331	single shaft, integral gearbox	41,730 → 2,000 rpm	20.9:1
This design	rear drive; offset + planetary	22,000 → 1,700 rpm	12.94:1

C. The T700-Class Core

In the T700 the power-turbine shaft runs through the gas generator to the front of the engine (Fig. 5), and both rotors run above their first bending critical speed, so rotordynamics matters from the start [11]. The engine has three oil sumps, an integral oil tank of about 6.9 L, a 3 µm filter, a chip detector and an emergency air-oil mist for its front sumps [12]. The CT7 shows the accessory split used later in this design: the starter-generator stays on the gas-generator

gearbox, while the propeller control unit (PCU), the overspeed governor (OSG), the hydraulic pump and the AC generator sit on the propeller gearbox and turn with the power turbine [3].

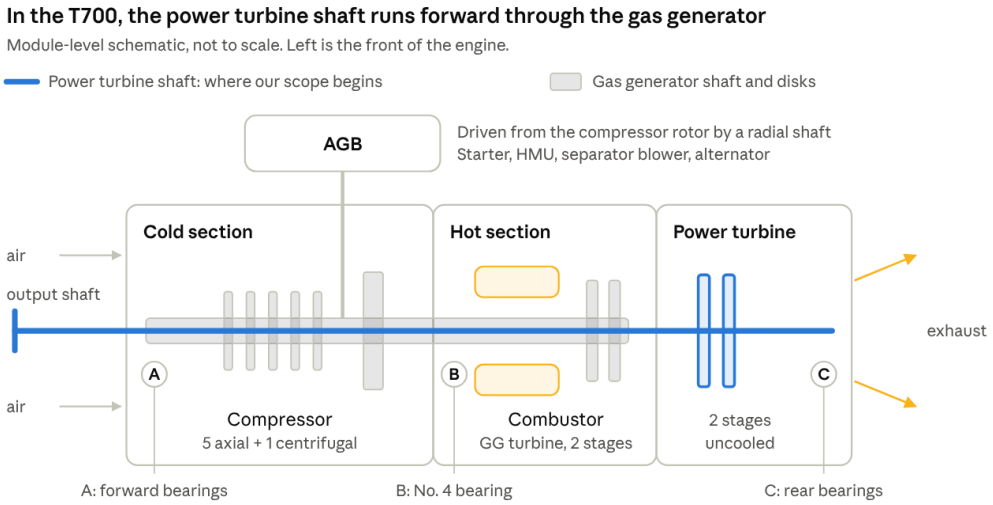


Fig. 5 Arrangement of the T700 core: the power-turbine shaft runs forward through the gas generator and leaves through the middle of the inlet.

D. Certification Basis

Before an engine may fly, an aviation authority must agree, clause by clause against a written code, that it is safe. The code does not say how to design a gearbox; it says what must never happen and what must be shown, and that is what shapes the design.

In a turboprop the gearbox, the shafts and the oil system are part of the engine type design, so the whole scope falls under the engine codes, EASA CS-E [13] and 14 CFR Part 33 [14]; both reference engines hold EASA type certificates, so these codes were taken as the design basis. STANAG 4671 [15] sets the aircraft level for the customer UAV. Fig. 6 maps the drive chain onto the clauses, and Table 4 lists the clauses that shaped the design most.

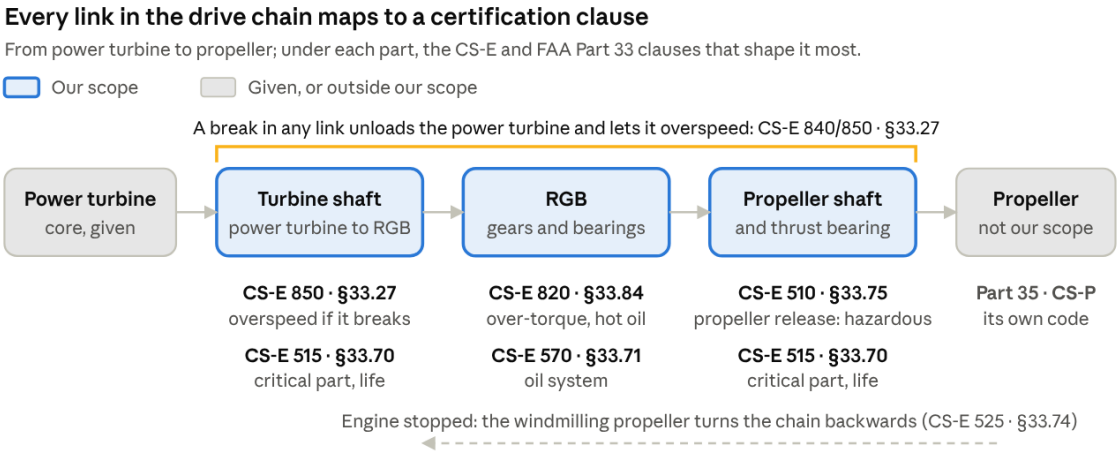


Fig. 6 Every link of the drive chain maps to a certification clause; a break anywhere between the power turbine and the propeller unloads the turbine and can let it overspeed.

Table 4 Certification clauses that shaped the design.

Topic	CS-E / Part 33	What it means for the transmission
Safety analysis	510 / § 33.75	release of the propeller is hazardous: at most 10^{-8} per flight hour from a single failure
Critical parts	515 / § 33.70	propeller shaft, input shaft, PT output shaft get approved lives and damage-tolerance assessments
Overspeed after shaft break	840, 850 / § 33.27	a break must be detected and the fuel cut, or the break shown to be extremely remote [16]
Over-torque	820 / § 33.84	15 minutes at the maximum torque with the gearbox oil at its hottest, then teardown
Vibration	650 / § 33.83	vibratory plus steady stresses below the endurance limits up to 103% speed
Endurance	740 / § 33.87, 33.93	150 h test, then every part must be fit for further service
Oil system	570 / § 33.71	filter with bypass, tank expansion space, chip detection
Continued rotation	525 / § 33.74	a windmilling propeller must not cause hazards without oil pressure

E. The Requirement Set

A requirement turns a wish into a sentence that can be checked: not "the gearbox should be light" but "the dry mass shall not exceed 110 kg". Each one also states where its number comes from and how it will be proven.

Requirements were built in three layers: eight customer needs (N), ten assumptions that turn them into numbers (A), and system requirements in eleven groups (system, loads, safety, dynamics, strength, lubrication, accessories, propeller interface, controls, maintainability, environment). Each requirement is a checkable "shall" sentence with a source, a verification method (analysis, test, inspection or demonstration), the parts it applies to and a status: Agreed, TBR or TBD. Aircraft-side loads used the pre-2017 values of 14 CFR 23.337 and 23.371 with the factor of safety of 23.2230 [17]. The first set had 53 requirements; through the reviews it grew to 69 (Fig. 7). Table 5 lists the values that size the transmission.

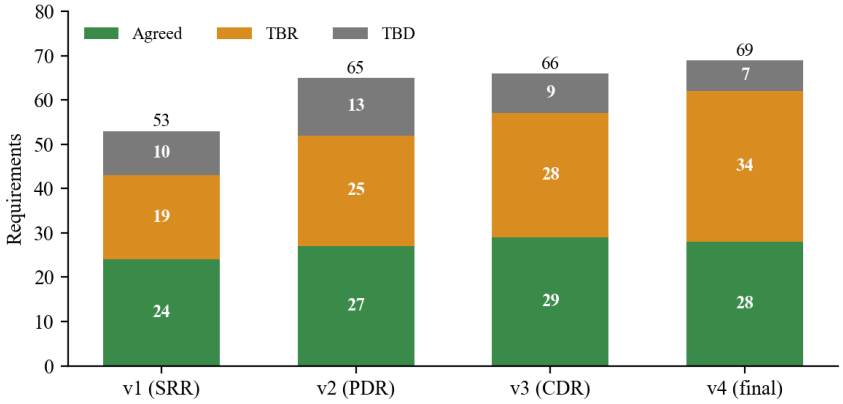


Fig. 7 Number and status of the requirements at each review. Green: agreed values; amber: provisional (TBR); grey: still unknown (TBD).

Table 5 Principal design values (requirements, final version).

Quantity	Value	Requirement
Take-off and maximum continuous power	895 kW, flat-rated to 53 °C	SYS-01, ENV-01
Speeds and ratio	22,000 → 1,700 rpm, 12.94:1 ± 0.5%	SYS-02
Propeller torque, continuous / transient (20 s)	5,027 / 6,915 N·m	SYS-01, LD-01
Steady speed range; transient overspeed	90–101%; 110% for 20 s	SYS-03, SYS-04
Propeller thrust, both directions	20 kN	LD-03, LD-04

Quantity	Value	Requirement
Gyroscopic moment (2.5 rad/s yaw + 1.0 rad/s pitch)	≈7.2 kN·m	LD-05
Efficiency at maximum continuous power	≥ 98.5% (≤ 13.6 kW of heat)	SYS-05
Dry mass of the RGB	≤ 110 kg (TBR)	SYS-06
Gear safety factors (ISO 6336)	$S_H \geq 1.2$, $S_F \geq 1.5$	STR-01
Bearing life	$L_{10m} \geq 12{,}000$ h	STR-02
Overhaul interval; service life	4,000 h; 20,000 h	STR-05
Running after loss of oil pressure	≥ 6 min at 75% power with an oil mist	LUB-07
Probability of propeller release	$\leq 10^{-8}$ per flight hour, single failure	SAF-01

Why the gyroscopic moment matters

A spinning propeller resists being turned, like a bicycle wheel. When the aircraft yaws or pitches, the propeller pushes back with a gyroscopic moment $M = I_p\,\Omega\,q$, where I_p is its polar inertia (15 kg·m^2), Ω its spin (178 rad/s) and q the turn rate. For the 2.5 rad/s yaw and 1.0 rad/s pitch of the design rules this is about 7.2 kN·m, more than the propeller torque itself, and it is what sizes the propeller shaft and its bearings.

IV. Concept Selection

Phase 1 answered two big questions: how to bring the power of a front-drive core to a rear propeller, and which gear arrangement to use. In short: the power-turbine shaft is extended out of the rear of the core (rear drive), and the gearbox is an offset helical first stage followed by one slow planetary stage.

A. Pusher Installation

The core breathes in at the front and blows out at the back, and its power shaft leaves at the front; the pusher propeller sits at the back. Something has to give: either the core is turned round, or the power is carried past it, or the shaft is led out through the hot end. These are the three layouts compared below.

Three layouts were compared (Fig. 8). In **A** the core is mounted backwards, so that its front drive meets the gearbox at the rear; the inlet then sits next to the propeller and the exhaust points forward. In **B** the core keeps its normal direction, and a front gearbox and a long shaft outside the core carry the power to the rear. In **C**, rear drive, the power-turbine shaft is extended aft through the exhaust, as GE did for the T700-701K helicopter engine [18]; the exhaust is split into two branches around the shaft.

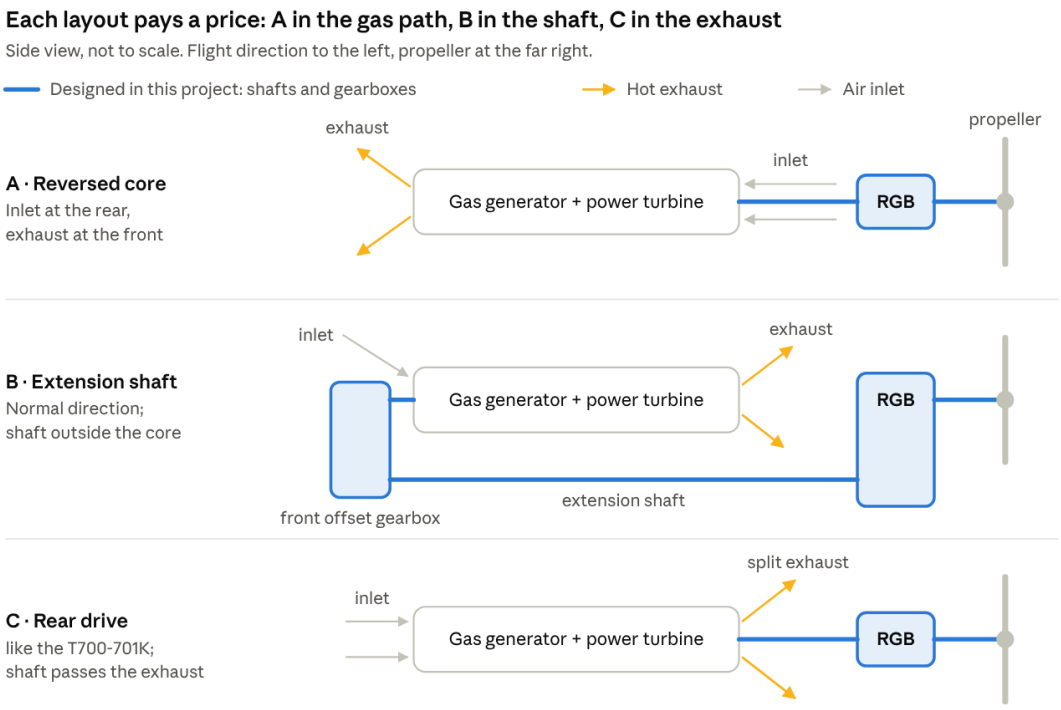


Fig. 8 The three installation options, side view, not to scale. Blue: parts designed in this project; orange: hot exhaust; light blue: inlet air.

Precedents exist for C: the Garrett TPF351-20 was built for the pusher Embraer/FMA CBA-123 Vector [19, 20], and AlliedSignal patents show the exhaust ducted around an aft gearbox and ejected around the propeller hub [21, 22]. Published data of the T700-701D and its rear-drive version -701K show what rear drive costs: 4–7% less power, 2.4–3.2% more specific fuel consumption, 21 kg and 44 cm more. A simple free-turbine cycle model quantified the effect of duct losses: 1% of inlet pressure loss costs about 1.7% of power and 0.7% of fuel.

Each option was scored from 1 to 5 on eight weighted criteria: installed performance, fit with the customer aircraft, mass and efficiency, rotordynamic risk, thermal environment, maintainability, certification risk and precedent. The totals were A 66, B 56 and C 67. Fig. 9 shows how the ranking depends on the weight given to the installation criteria. B was dropped: its second gearbox and long shaft add mass, critical parts and certification risk. C was chosen over A because it keeps the inlet forward and the exhaust ahead of the propeller, as in today's PT6A-67A installation, has two precedents, and costs about 2–3% in fuel, 35–55 minutes on a 30-hour mission, well inside the 35% power margin of the core.

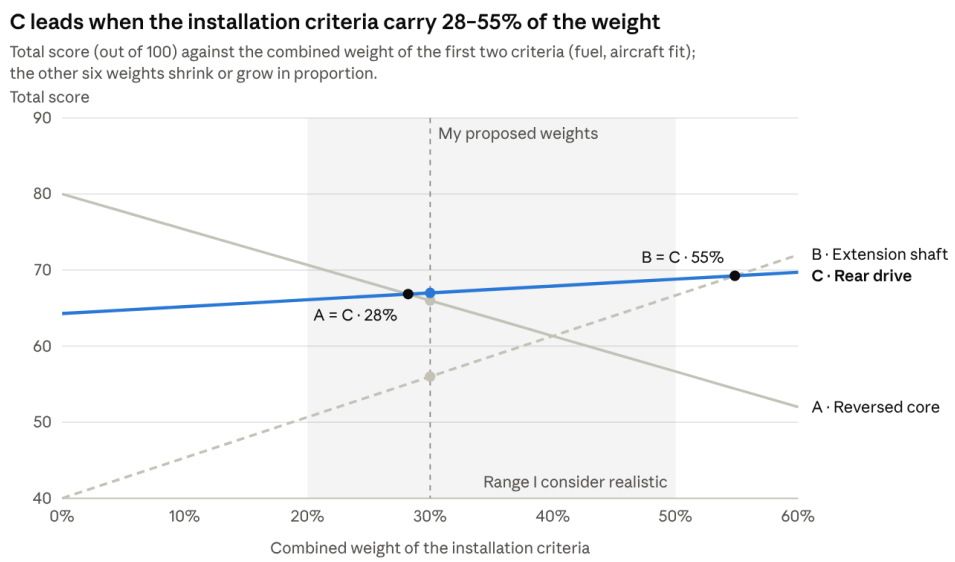


Fig. 9 Sensitivity of the installation ranking to the combined weight of the two installation criteria. C leads in the realistic range of 28–55%.

B. Gearbox Architecture

Two stages of about 3.6:1 are needed, and each stage can be a pair of gears side by side or a planetary set, so there are only a few natural combinations. The choice is not only about mass: how fast each part spins decides which bearings can be used, and that in turn decides what happens when the oil runs out.

Three two-stage arrangements were sized (Fig. 10): **P**, two coaxial planetary stages as on the PT6A; **O**, an offset helical first stage followed by a planetary stage [6]; and **S**, an offset split-path train [23, 24]. P is the lightest (about 27 kg of gears against 34 kg for O and 44 kg for S) and keeps the propeller on the engine axis.

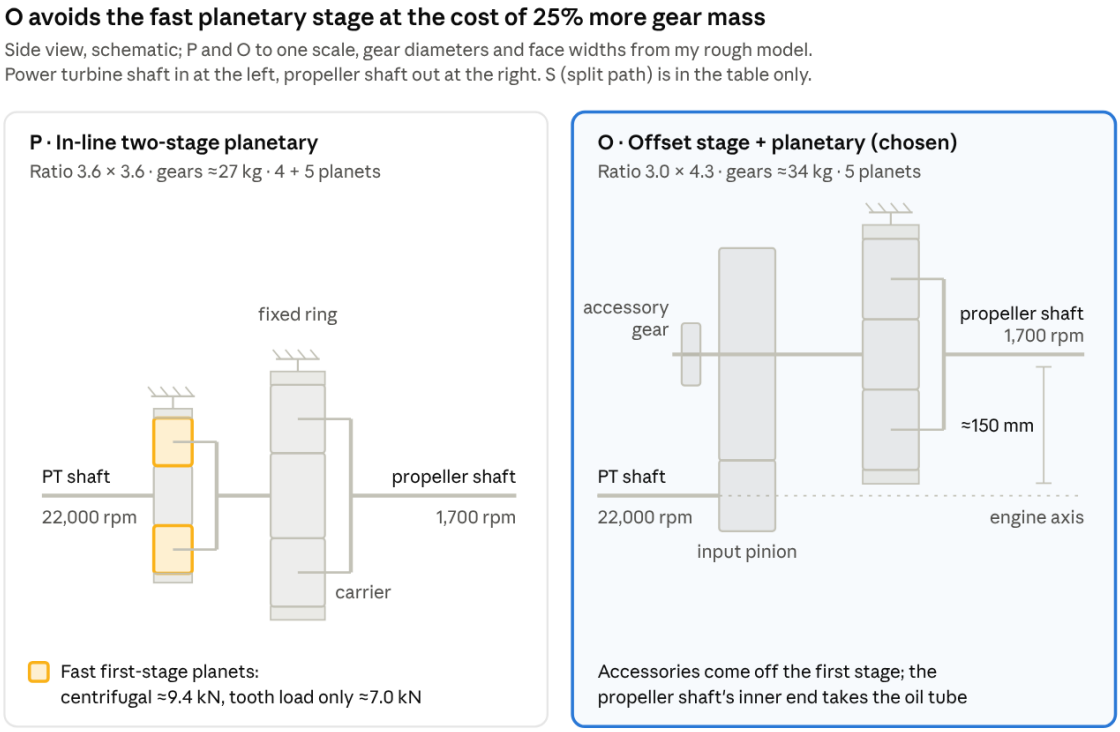


Fig. 10 Arrangements P (in-line, two planetary stages) and O (offset first stage + planetary stage), side view at one scale. In P the first planetary stage turns fast; in O the only planetary stage turns slowly.

Key idea: why the in-line planetary layout was rejected

In arrangement P the first carrier turns at about 6,100 rpm. At that speed each planet is pushed outward by a centrifugal force of about 9.4 kN, more than the 7.0 kN it takes from the teeth. Rolling bearings small enough to fit inside the planets cannot carry this for long, which is why gearboxes of this type use journal (oil-film) bearings under the planets [25]. Journal bearings fail within seconds without oil, so P would make it impossible to keep running after an oil loss, a requirement of the military customer. In arrangement O the planetary stage turns slowly, and rolling planet bearings meet their life.

Arrangement O also brings the interfaces for free: the accessory pads are driven from the first-stage gear, the pitch-change oil for the propeller enters at the free inner end of the propeller shaft, and the opposite propeller rotation, if an aircraft needs it, is obtained by adding two idler gears with most parts unchanged. The Allison T56 uses the same arrangement at almost the same ratio, 13.54:1 [26, 27]. The price is about 7 kg more gear mass and a propeller axis about 150 mm away from the engine axis. The weighted scores were O 78, P 75 and S 66 (Table 6), and O was chosen.

Table 6 Scoring of the gearbox arrangements (1 poor – 5 good; total = weighted average × 20).

Criterion	Weight, %	P	O	S
Mass	20	5	4	2
Bearing life and reliability	20	4	4	5
Running through oil interruption and negative g	10	1	3	4
Fit with installation C and the aircraft	15	5	3	2
Accessory pads, pitch oil, opposite hand	15	2	5	4
Development risk and precedent	10	5	4	3
Maintenance and part count	10	3	4	3

Criterion	Weight, %	P	O	S
Total	100	75	78	66

C. Preliminary Sizing and the Number of Planets

Once the arrangement is fixed, the design becomes a puzzle of whole numbers: tooth counts that give the right ratio, planets that fit around the sun, gears just large enough for their load. A computer can try every combination; the engineer's job is to write the rules it must obey and to check that its answers make sense.

A MATLAB tool searched every combination of tooth numbers for arrangement O. It chooses three things itself: the tooth numbers, the number of planets (3, 4 or 5) and the module, which sets the size of a tooth (pitch diameter = number of teeth × module); it takes the smallest standard module between 2 and 8 mm that passes every check. The tooth numbers must also obey four rules, each with a physical reason (Table 7).

Table 7 Rules a valid set of tooth numbers must satisfy, and why.

Rule	Condition	Reason
Ratio	$(z_2/z_1)(1 + z_r/z_s)$ within ±0.5% of 12.941	whole numbers of teeth cannot give the exact ratio; ±0.5% is about ±8.5 rpm at the propeller
Hunting teeth	meshing tooth numbers share no common factor (required in the first stage, preferred in the planetary stage)	every tooth meets every tooth of its partner, so a small defect does not keep wearing the same few teeth
Assembly	$z_r = z_s + 2z_p$, and $(z_s + z_r)/N$ is a whole number	evenly spaced planets must engage the sun and the ring at the same time, or the last planet cannot be fitted
Space	at least 15 mm between the tips of neighbouring planets	room for an 11 mm carrier post with 2 mm clearance on each side

A design was accepted if its teeth reached $S_H \geq 1.2$ and $S_F \geq 1.5$ at maximum continuous torque for 10^{10} load cycles and survived the transient torque statically, if its planet bearing reached $L_{10} \geq 12{,}000$ h, if its shafts had infinite fatigue life (Goodman safety ≥ 1.5), if its critical speeds stayed 20% away from the running speeds and if the gearbox weighed at most 100 kg. Each candidate was sized with a simplified ISO 6336:2019 gear rating [28], a planet-bearing model after ISO/TR 1281-1 [29], shaft fatigue checks after Shigley [30] and a part-by-part mass estimate. Of 6,367 tooth sets, 5,733 gave valid designs; the full search takes about 8 s. Fig. 11 shows the lightest designs against the first-stage ratio.

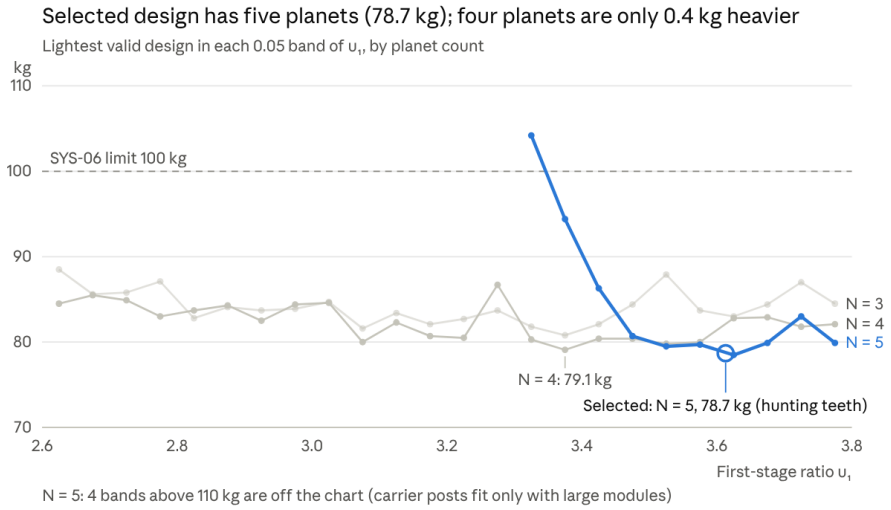


Fig. 11 Lightest valid design for each first-stage ratio and planet count. Five planets only compete above a first-stage ratio of about 3.45; four planets stay at 79–83 kg over a wide range. The lightest four-planet design is only 0.4 kg heavier than the five-planet one; the four-planet set with hunting teeth chosen after the PDR is 1.1 kg heavier.

Why the application factor K_A was raised from 1.1 to 1.25

The first full search found a 73.3 kg design. Every coefficient was then compared clause by clause with ISO 6336:2019, which brought five corrections; the application factor K_A, which covers load peaks above the nominal torque, mattered most (about 5 kg). The first value, 1.1, is the table entry for a slightly shocky driver turning a uniform load. Here the case is the reverse: a uniform driver (a gas turbine burns continuously, without firing pulses) turns a load with light shocks (a pusher propeller runs in the wake of the wing and fuselage, so each blade meets the uneven flow once per turn). The table then gives 1.25, which is kept until a measured or calculated torque spectrum exists (requirement STR-01).

How robust is the result? Table 8 repeats the full search with the rating assumptions changed one at a time and all together. The mass of the selected design changes by at most 6.2 kg, so even the worst combination stays 15 kg below the 100 kg limit of the time; the tooth numbers, however, change from case to case, which is why they were fixed only by the detailed rating of Phase 2. The architecture choice survives as well: the mass score of O stays at 4, so the 78 against 75 result of Table 6 does not change.

Table 8 Sensitivity of the lightest design's mass to the rating assumptions (full search repeated for each case; f_{ZCa} is the flank-modification factor).

Case	Mass, kg	Change, kg
Base: K_A 1.25, K_v 1.1, f_{ZCa} 1.0	78.7	—
K_A 1.35 (gas turbine also counted as slightly shocky)	79.9	+1.2
K_v 1.2	81.1	+2.4
f_{ZCa} 1.07 (flank modification from manufacturer experience)	79.3	+0.6
f_{ZCa} 1.2 (no flank modification)	81.1	+2.4
K_v 1.2 and f_{ZCa} 1.07	81.5	+2.8
K_A 1.35, K_v 1.2 and f_{ZCa} 1.2 together	84.9	+6.2
K_A 1.1, the value before the standards check	73.6	−5.1

The standards check also showed that some of the values used are valid only under certain conditions, and these became requirements at the PDR (STR-06 to STR-08):

- optimised profile and helix modifications on the first stage, assumed by f_{ZCa} = 1.0 (without them 1.2, about +2.4 kg);
- a core hardness of at least 30 HRC for the carburized gears, on which their tooth-root strength depends;
- an aluminium-free nitriding steel for the ring with at least 650 HV, because nitriding steels with aluminium are allowed a much lower root stress;
- a ring rim of at least 3.5 modules, below which the ring flexes and its root stress rises;
- a floating sun and a flexible ring, which the load-sharing factor assumes.

The lightest design had five planets (78.7 kg). The lightest four-planet design was only 0.4 kg heavier, and with hunting teeth required in the planetary stage too, the four-planet set 31/110 | 45/37/119 was 1.1 kg heavier. Table 9 compares the two at the PDR, after the accessory masses of the next step had been added. A difference of one kilogram is within the accuracy of the rough mass model, whereas the 18 mm gap between five planets is very close to the unverified 15 mm floor. The four-planet design also shares the load better [31] and leaves three times the margin on the losses, so it was chosen after the PDR.

Table 9 Four or five planets: the two candidate designs at the PDR.

Quantity	Four planets	Five planets
Tooth numbers	31/110 | 45/37/119	31/112 | 39/31/101
Gearbox mass at the PDR	77.6 kg	76.6 kg
Gap for the carrier posts (at least 15 mm)	38 mm	18 mm
Load-sharing factor K_γ (smaller is better)	1.15	1.19

Quantity	Four planets	Five planets
Margin to the 13.6 kW loss limit	0.84 kW	0.26 kW
Uncertainty cases above the loss limit	1 (by 0.09 kW)	3
Distance of the propeller axis from the PT axis	146 mm	148 mm

The resulting power path is shown in Fig. 12, and the overall ratio is

$$i = (z_2/z_1)\,(1 + z_r/z_s) = (110/31)(1 + 119/45) = 12.932 \quad (3)$$

which is 0.07% below the nominal value and inside the ±0.5% tolerance.

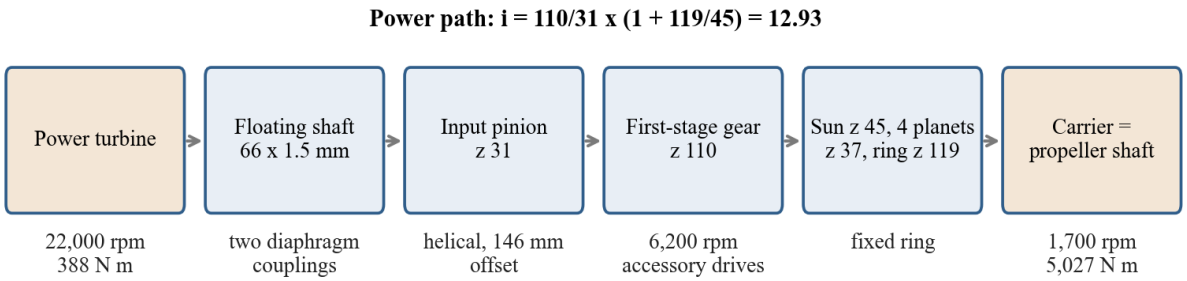


Fig. 12 Power path of the selected design with the shaft speeds and torques at full power.

D. Accessories, Losses and the Oil System

Besides turning the propeller, the gearbox runs the engine's helpers: the generator, the governor that caps the propeller speed, the pitch-control unit and its own oil pumps. And no gearbox is perfect: a small share of the power turns into heat in the meshes and bearings, and the oil has to carry it away.

A separate oil system for the gearbox, as on the CT7-9, scored 87 points against 60 for a hybrid and 53 for a system shared with the core, so the core's oil system stays unchanged. The four accessories driven by the power turbine, the AC generator, the overspeed governor, the propeller control unit and the gearbox oil pumps, take their power from the intermediate shaft (Fig. 13, left). A loss model gave 12.76 kW at full power, an efficiency of 98.67%, below the 13.6 kW limit; the gear meshes, the planetary stage and the bearings make most of it (Fig. 13, right).

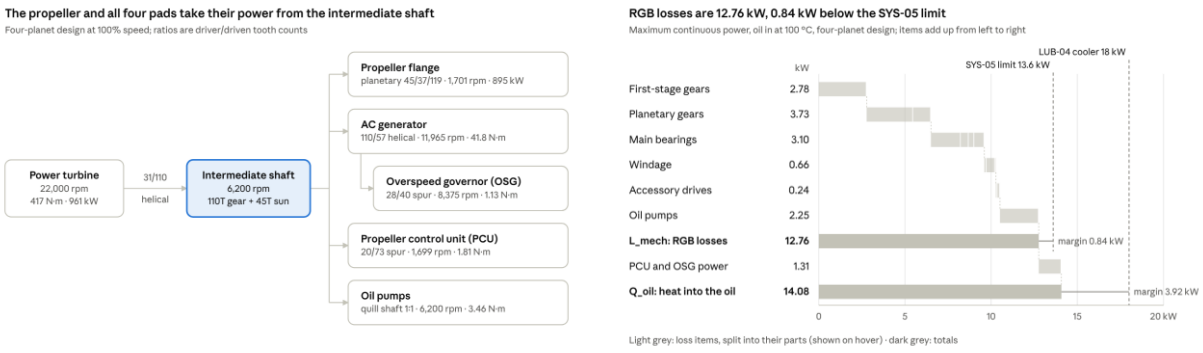


Fig. 13 Left: the four accessory pads, all driven from the intermediate shaft. Right: breakdown of the gearbox losses at maximum continuous power (12.76 kW at the PDR).

Three interface choices were also made here. The propeller turns opposite to the power turbine: the external first-stage mesh reverses the direction, and with the ring fixed the carrier turns the same way as the sun. If the aircraft needs the other direction, two idler gears in the first stage provide it without touching the planetary stage, at the cost of a

changed offset and idler teeth loaded on both flanks (SYS-07, open until the aircraft maker decides). The 895 kW are defined without accessory load, and the torque limit acts on the power-turbine shaft, as in the PT6A-67A data sheet (SYS-08); the losses were still computed for the heavier case with the accessories loaded. Finally, Table 10 lists the sensors that the engine control reads from the gearbox. Two of them need a word: a phase-shift torque meter measures the small twist of a shaft between two toothed wheels, and a phonic wheel gives one pulse per tooth to a magnetic pickup, so that the pulse rate gives the speed.

Table 10 Sensors on the gearbox and what each is for.

Sensor	Where	Purpose
Torque meter (phase shift)	power-turbine shaft at the gearbox input	engine torque; the power-turbine speed comes from the same signal (CTL-01)
Power-turbine speed × 2	power-turbine shaft	one for control, one for the overspeed and shaft-break protection (CTL-02)
Propeller speed × 2	phonic wheel on the propeller shaft	shaft-break detection: if the chain breaks, the ratio of the two speeds changes (SAF-03)
Oil pressure	after the filter	oil reaches the gears and bearings (LUB-03)
Oil temperature	gearbox oil inlet	the cooler is sufficient (LUB-02)
Filter clogging	filter	warns when the pressure drop across the filter grows (LUB-05)
Chip detectors × 2	scavenge lines	magnets catch metal particles: early warning of wear (LUB-05)
Oil debris monitor	scavenge line	counts metal particles continuously (CTL-03)
Tank level	oil tank	oil loss (LUB-06)
Accelerometers × 3	case wall over the ring gear, outer propeller bearing boss, first-stage case	condition monitoring (CTL-03, Section IX)

E. Shaft-Break Protection

A power turbine that loses its load behaves like a car engine at full throttle whose clutch is suddenly released, except that it reaches a destructive speed within a fraction of a second. The protection must notice the break and cut the fuel faster than the rotor can run away.

If the drive chain breaks anywhere between the power turbine and the propeller, the turbine suddenly loses its load and accelerates. The chosen protection compares two independent speed signals, detects the break within 14.5 ms and cuts the fuel 25 ms later. Fig. 14 shows that the peak speed then depends mostly on how fast the core's torque decays after the fuel is cut: with a time constant $\tau = 0.1$ s the peak is 123.3%, which became a requirement on the core (SAF-06).

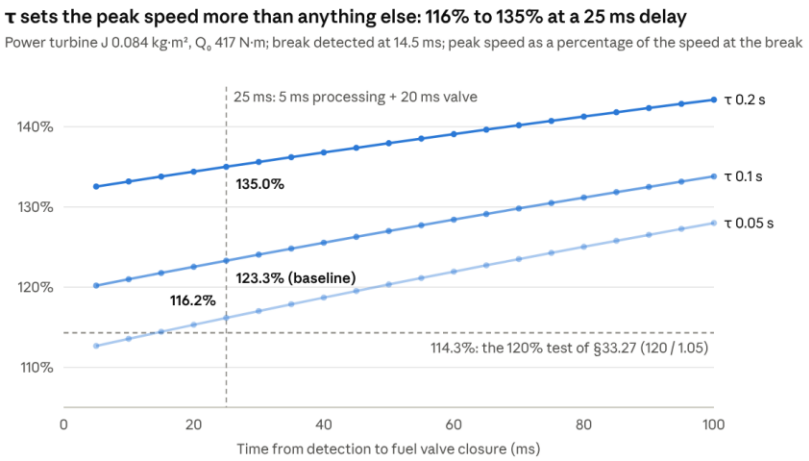


Fig. 14 Peak power-turbine speed after a shaft break against the time from detection to fuel cut, for three decay constants of the core torque.

F. The Risk Carried into Detail Design

A spinning shaft can also whirl: its centre traces a small circle, like a skipping rope, either in the direction of rotation or against it.

A rotordynamic model of the propeller shaft in the open-source code ROSS [32] found no forward-whirl critical speed but a backward-whirl crossing just below the steady speed range (Fig. 15). In backward whirl the shaft centre orbits against the direction of rotation; perfectly round supports cannot excite it, but supports that are stiffer in one direction than the other can. This was the only risk rated high at the PDR (Fig. 16) and the first task of Phase 2. The RGB was estimated at 77.6 kg at the PDR, and the requirements grew to 65.

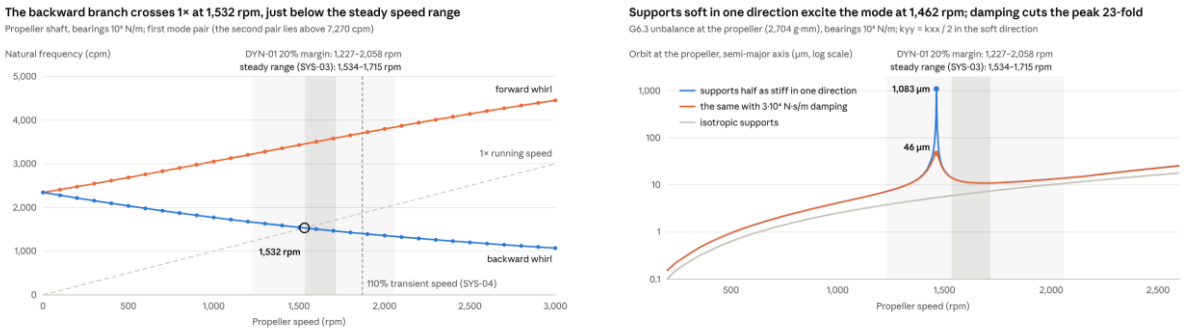


Fig. 15 Left: Campbell diagram of the Phase 1 propeller shaft; the backward branch crosses the 1× line at 1,532 rpm, just below the steady range. Right: unbalance response with supports soft in one direction; the crossing appears as a resonance.

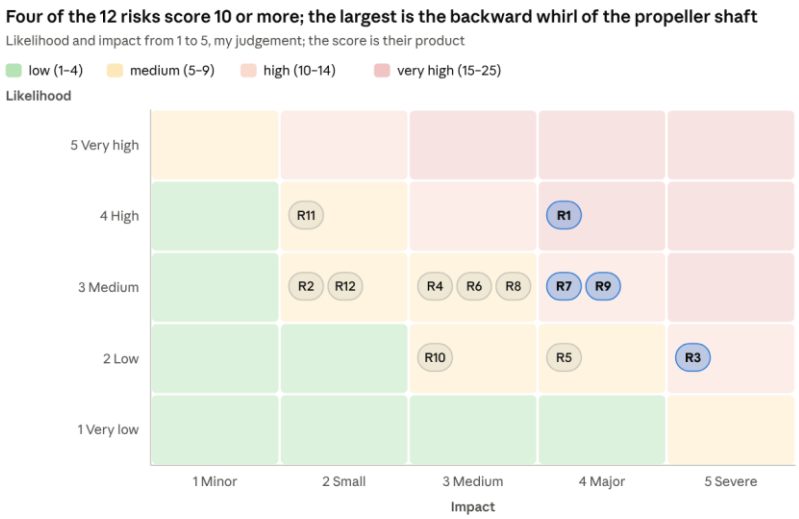


Fig. 16 Risk matrix at the PDR (likelihood against impact). The largest risk, R1, is the backward whirl of the propeller shaft.

V. Detail Design

Phase 2 took the concept down to its parts in eleven steps, each with its own calculation tool, tests and report. The steps were not independent: a stiffer shaft chosen for one reason changed a torsional mode, a wider gear changed a bearing, a new case changed a stiffness. This section follows the power from the propeller back to the case and shows, for each part, the problem found, the fix and the resulting margin. The main gear data are in Table 11 and the CAD model in Fig. 17.

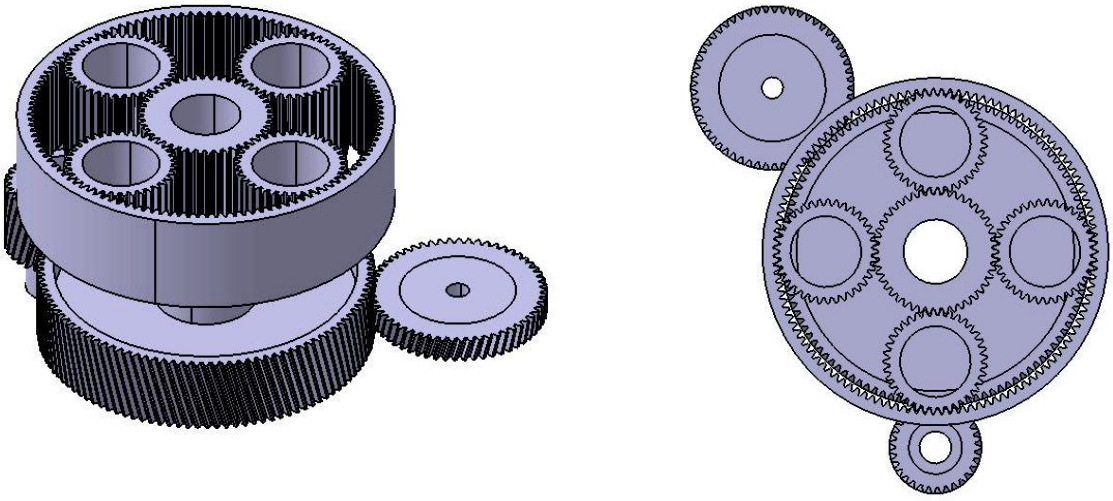


Fig. 17 The gear train modelled in CATIA V5 with true involute teeth: input pinion, first-stage gear with the generator drive, sun, four planets, fixed ring and carrier with the propeller shaft. Left: three-quarter view; right: view along the axis.

Table 11 Main gear data of the detailed design.

Quantity	First stage (helical)	Planetary stage (spur)
Teeth	z_1 31 / z_2 110	z_s 45 / z_p 37 / z_r 119, N = 4
Speeds at 100%	22,000 / 6,200 rpm	carrier 1,701 rpm, planet 5,471 rpm
Offset / centre distance	146.0 mm (PT axis below)	a = 82 mm
Face width; planet rim	—	70 mm; 8 mm (4 modules)

Quantity	First stage (helical)	Planetary stage (spur)
Material	Pyrowear 53, carburized, superfinished	Pyrowear 53; ring nitrided 32CrMoV13
Lowest S_H / S_F (with dynamic load)	1.53 / 2.43	1.28 / 1.55
Load sharing K_γ (P99)	—	1.257 with drawing tolerances (limit 1.35)

A. Propeller Shaft: Removing the Backward-Whirl Risk

Picture the propeller as a heavy disc at the end of a rod. When the rod flexes, the disc tilts; because the disc spins, it acts like a gyroscope and fights the tilt. The tilting vibration of the shaft therefore splits into two forms: one that circles with the rotation and stiffens as the speed rises, and one that circles against it and softens. The softening one is the danger, because it can sink into the running speed range.

The propeller's polar moment of inertia is twice its diametral one (I_p = 15, I_d = 7.5 kg·m^2). When the shaft spins, this gyroscopic effect splits the shaft's conical (tilting) mode into a forward branch that rises with speed and a backward branch that falls (Fig. 18, left). A twelve-element Timoshenko beam model of the shaft with its bearings, centring springs and housing was built in MATLAB and checked against the ROSS model to 0.008%. On the Phase 1 shaft (75/45 mm) the backward branch crossed the 1× line at 1,443–1,636 rpm, inside or just below the steady range; with a housing half as stiff in one direction the propeller orbit reached 1.08 mm.

Four cures were compared. A squeeze-film damper, the PDR proposal, did not work, because the steady propeller loads push its journal out of the oil clearance. The chosen cure is simpler: a thin-walled 120/100 mm steel shaft, stiff enough to move the backward crossing to 2,065–2,631 rpm, above the 110% speed of 1,870 rpm (Fig. 18). The response ratio falls to 1.67 against a limit of 2.5, on one condition handed to the case design: each bearing seat must give at least 5×10^7 N/m in every direction.

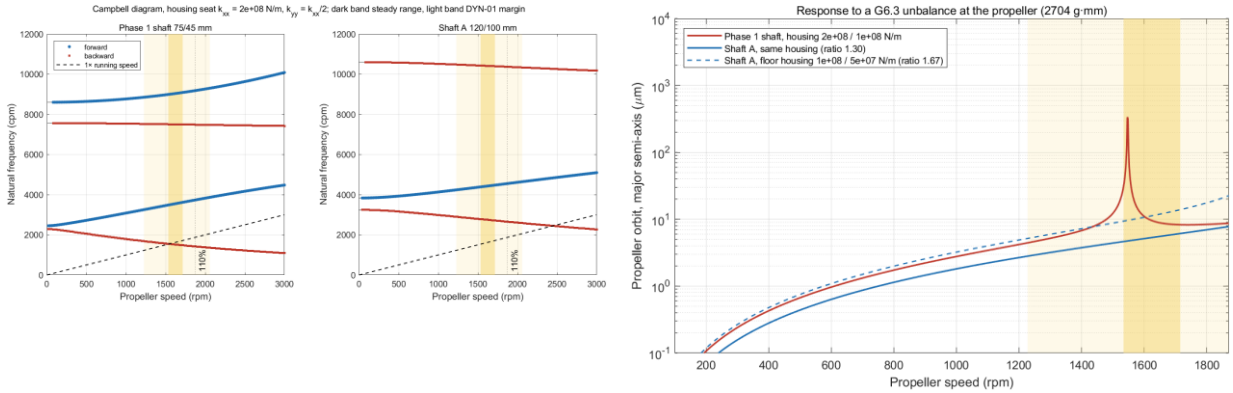


Fig. 18 Left: Campbell diagrams of the Phase 1 shaft and of the chosen 120/100 mm shaft; the backward branch (red) leaves the steady band. Right: propeller orbit under a standard unbalance; the resonance of the old shaft disappears from the operating range.

> **Term: Campbell diagram**
>
> A Campbell diagram plots the natural frequencies of a rotating part against its speed. Straight lines through the origin show the excitations, for example once per revolution (1×). Wherever a frequency curve crosses an excitation line inside the operating speed range, the part can resonate. Designers either move the crossing out of the range, as here, or add damping.

B. Power-Turbine Output Shaft and Its Tunnel

The power has to travel half a metre from the turbine to the gearbox through the hottest part of the engine, the exhaust collector. The shaft that carries it is a thin tube spinning at 22,000 rpm: stiff enough not to whip, flexible enough at its ends to follow the core as it grows with heat, cool enough to keep its strength, and it also serves as the engine's torque meter.

In installation C the power leaves the core at the rear. A 0.55 m floating shaft with a flexible diaphragm coupling at each end carries it through a tunnel in the exhaust collector to the input pinion. The couplings absorb about 1.5 mm of thermal growth of the core and 0.25° of continuous misalignment by bending, so there is no spline to wear. The design space (Fig. 19, left) is bounded by the critical speed, the torsional stiffness and the twist the torque meter needs; the chosen AISI 4340 tube of 66 × 1.5 mm weighs 1.31 kg, has its first bending critical speed at 34,443 rpm and a fatigue safety of 7.32, and doubles as the torque-meter shaft (0.40° of twist at full torque, 0.66% resolution). In the tunnel, 15 mm of insulation and 7.5 g/s of buffer air keep the shaft below 124.5 °C in 700 °C exhaust gas (Fig. 19, right).

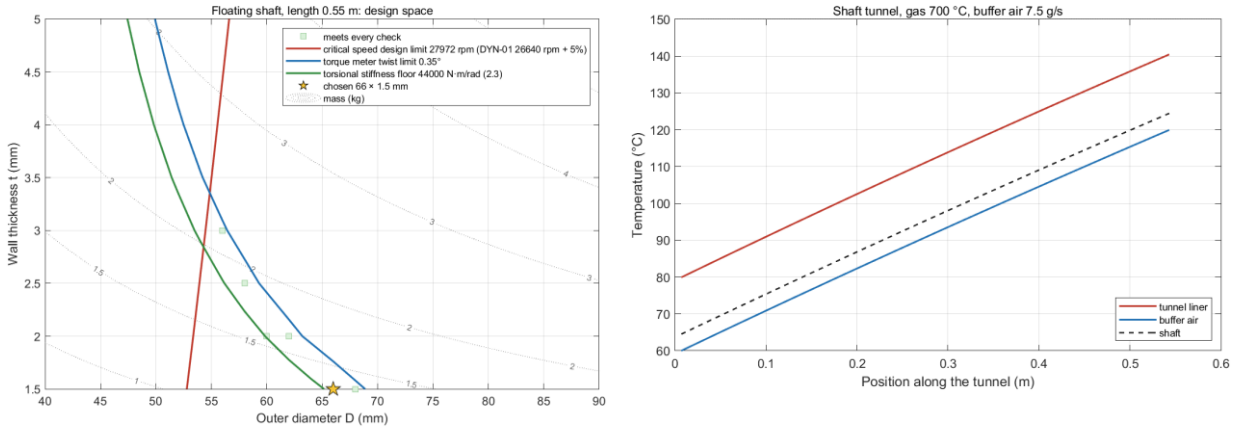


Fig. 19 Left: design space of the floating shaft (outer diameter against wall thickness) with the critical-speed, torque-meter and torsional-stiffness limits; the star is the chosen 66 × 1.5 mm tube. Right: temperatures along the tunnel in 700 °C gas with 7.5 g/s of buffer air.

C. Torsional Vibration and Sudden Stoppage

The drive chain is a row of heavy rotating masses, the turbine, the gears and the propeller, joined by shafts that twist slightly like springs. Such a chain has its own twisting rhythms, and every rotating part plucks it at its own frequencies; when a pluck matches a rhythm, the twisting grows. The same model also tells what happens when one end of the chain stops suddenly.

A 14-degree-of-freedom torsional model, with the planetary stage modelled after Lin and Parker [33], first showed that the torsional mode of a lighter 54 mm floating shaft sat at 343 Hz, inside the band swept by the power-turbine speed (330–370 Hz). The 66 mm shaft moved it to 414 Hz; all modes now stay at least 11.7% away from every excitation order (Fig. 20, left).

The same model answers a harsher question: what happens if the power turbine seizes suddenly while the heavy propeller keeps turning? The propeller shaft torque then grows as the stopping time shrinks, with a dynamic amplification of about 2 (Fig. 20, right). The gears can take 24,345 N·m at the propeller, so the core must take at least 0.22 s to stop; this became an interface requirement on the core maker (LD-08).

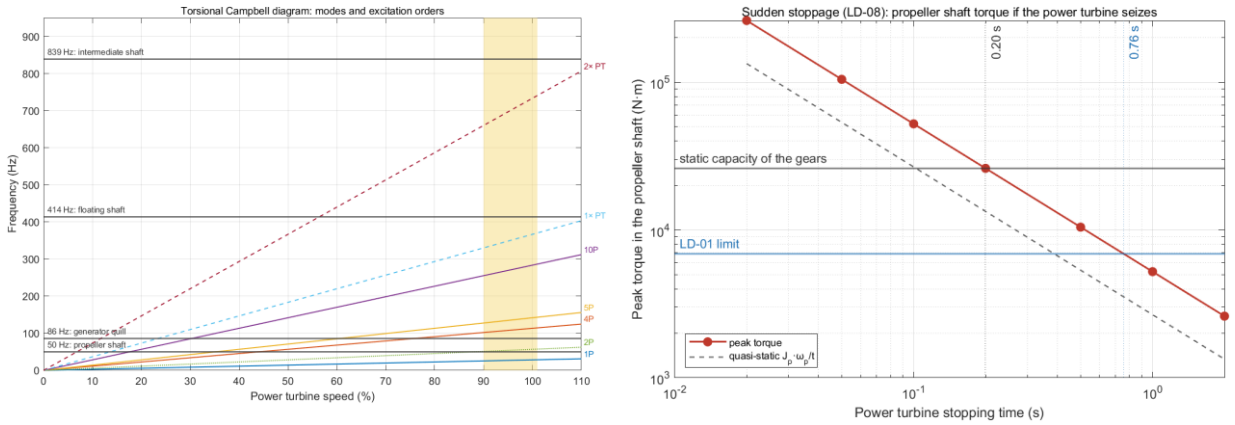


Fig. 20 Left: torsional Campbell diagram of the drive chain with the excitation orders; no mode crosses an order in the steady band. Right: peak propeller-shaft torque against the stopping time of the power turbine; below 0.22 s the static capacity of the gears is exceeded.

D. Gear Rating

Each time two teeth meet, they press on each other over a contact strip narrower than a millimetre at pressures of the order of a gigapascal, while the root of each tooth bends like a short cantilever. Repeated billions of times over the life, both can fatigue.

Gears fail in two main ways: the flanks pit (small pieces break out of the surface by contact fatigue) and the teeth crack at their root (bending fatigue). ISO 6336:2019 Method B [28] gives a safety factor against each; the requirement is at least 1.2 against pitting and 1.5 at the root. Fig. 21 shows the result for the final design. Three changes were needed to get there: superfinishing of all flanks to Rz 0.8 µm (with ground flanks only, the pitting safety would be 1.19), ring-gear tips shortened by 0.1 module so that they clear the start of the planet involute, and helix corrections with thicker carrier posts.

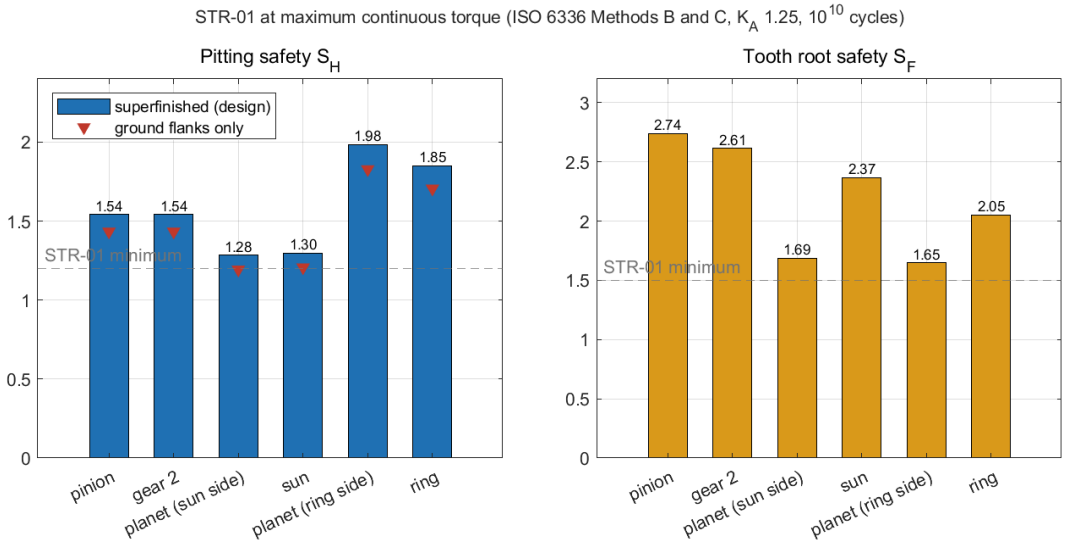


Fig. 21 Pitting safety S_H (left) and tooth-root safety S_F (right) of every gear at maximum continuous torque; the dashed lines are the required 1.2 and 1.5. Superfinishing raises the pitting safety of the planetary meshes above the line.

In a planetary stage the planets never share the load exactly equally, because of manufacturing errors. A Monte Carlo model of the stage with a floating sun gave a load-sharing factor K_γ of 1.32 at the 99th percentile (Fig. 22, right), larger than the 1.15 assumed in Phase 1; all gear and bearing ratings use it. Four ways to restore the margin were compared (Fig. 22, left), and superfinishing with a stiff planet bearing was chosen. The gears are carburized Pyrowear 53 (AMS 6308), because their bulk temperatures of 145–171 °C exceed the tempering limit of the usual AISI 9310; the ring is nitrided 32CrMoV13. Scuffing, the welding and tearing of flanks when the oil film breaks down at high temperature, was checked by the flash and integral temperature methods: with MIL-PRF-23699 oil at 132 °C the safety is at least 1.61 (Fig. 23), while the cold-climate oil MIL-PRF-7808 left no margin, which later led to a single-oil decision.

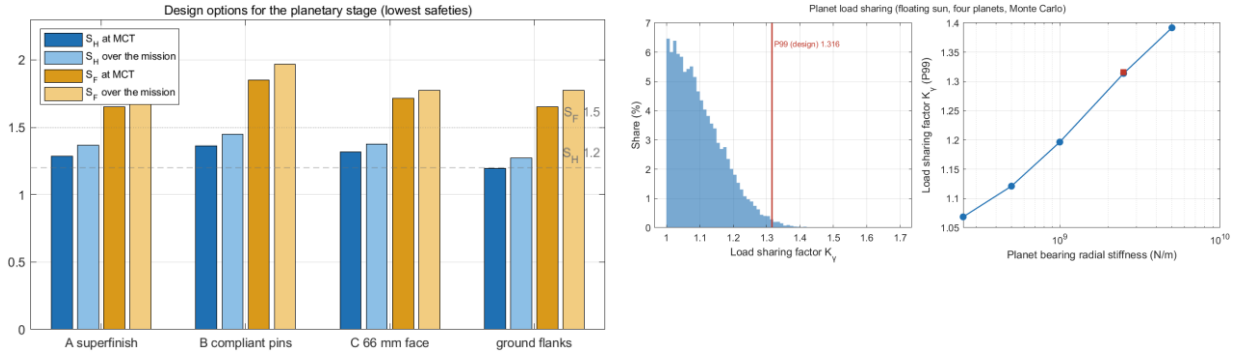


Fig. 22 Left: lowest gear safeties of the four options for the planetary stage (A, superfinishing, was chosen). Right: Monte Carlo distribution of the planet load-sharing factor and its sensitivity to the planet-bearing stiffness.

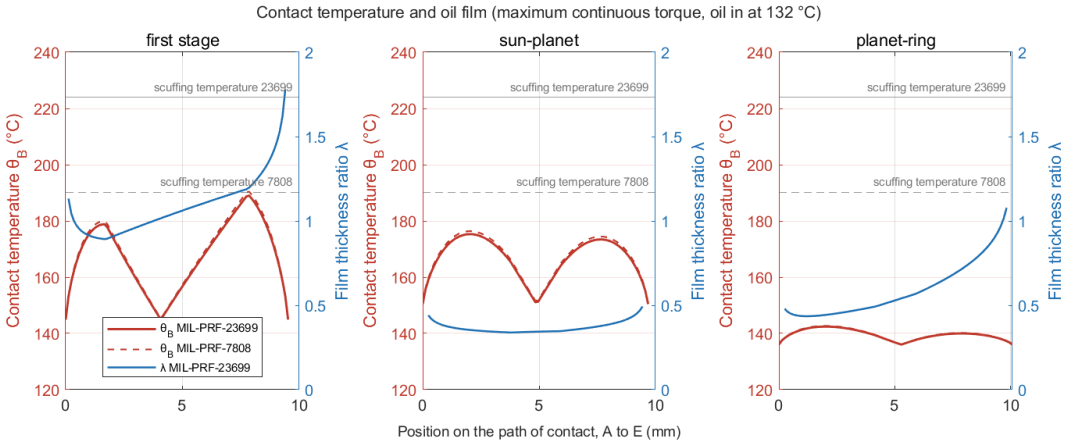


Fig. 23 Contact temperature along the path of contact of the three meshes at maximum continuous power with 132 °C oil, against the scuffing limits.

E. Bearings, Shafts and Propeller Retention

A rolling bearing carries its load on a handful of rollers at any moment, and every roller that passes a point of the raceway loads that point once more. After many millions of passes the steel tires just below the surface and a flake breaks out. That is why bearing life is a statistic: L_{10} is the time that 90% of identical bearings survive.

Every bearing must reach a modified rating life of at least 12,000 h (three overhaul intervals) over the mission load spectrum [34]. The planet bearing is the hardest case, because it must fit inside the planet gear and also carries the planet's centrifugal force. It first came out at only 3,600 h. It reached 14,500 h after the planetary face width was raised from 60 to 70 mm, the planet rim from 7.5 to 8 mm, and the bearing changed to one row of 18 rollers of 7.5 × 56 mm in full complement, without a cage (Fig. 24). The four-point contact bearing that takes the propeller thrust reaches 35,000 h. The intermediate shaft grew to 50/30 mm and the crowned spline that drives the floating sun to 56 mm; every shaft and spline meets its fatigue requirement (Fig. 25).

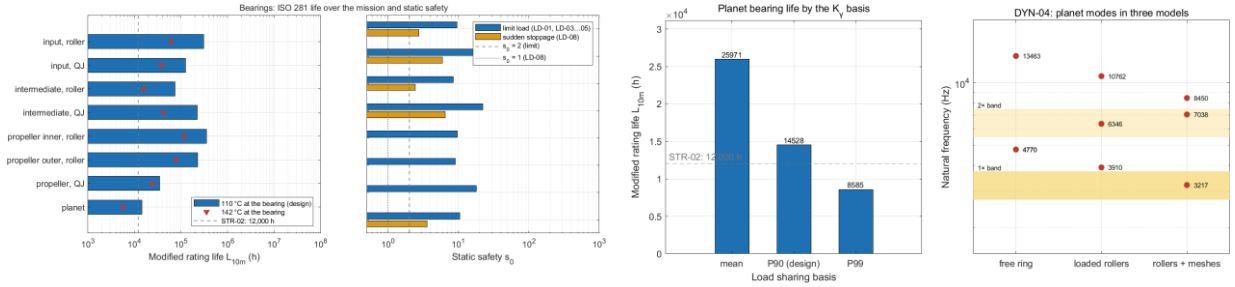


Fig. 24 Left: modified rating lives and static safeties of all bearings. Right: planet-bearing life for three bases of the load-sharing factor (mean, P90 used for design, P99) and the planet natural frequencies in three models.

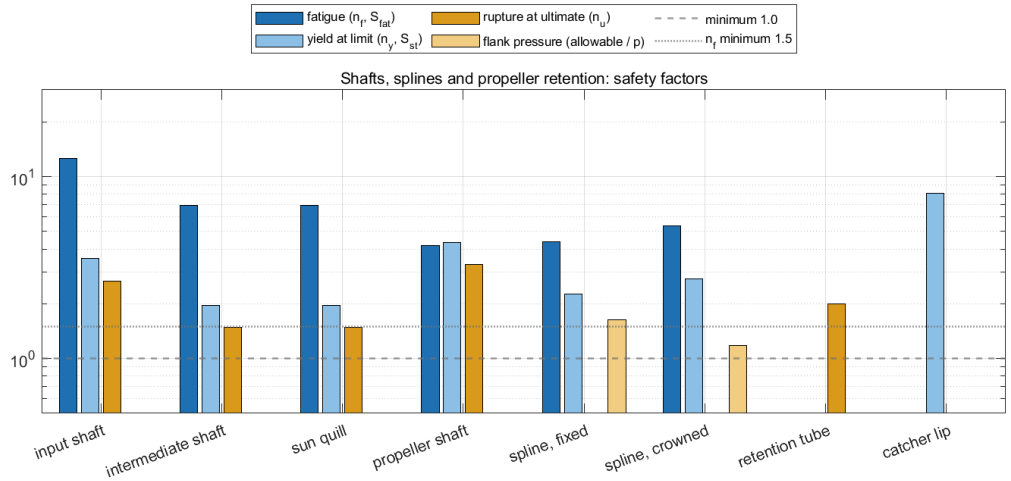


Fig. 25 Fatigue (Goodman) and static safety factors of the shafts, splines and retention parts; all lie above 1.

A pusher propeller normally pushes the shaft into the gearbox, but in windmilling or at low power it pulls outward. Release of the propeller is a hazardous engine effect, so a second load path was added: a Ti-6Al-4V retention tube of 90/80 × 600 mm inside the propeller shaft (+3.6 kg). In Phase 3 it was completed by a nose cover with a titanium catcher lip and a standby roller-bearing path (Section VII).

F. Accessory Drives

Each accessory is driven through a slim shaft with a deliberately weak neck, like a fuse in an electric circuit: if an accessory seizes, the neck breaks and the gearbox runs on unharmed.

The generator, overspeed governor, propeller control unit and oil-pump pads were rated in three load cases with the same gear tools; the lowest pitting safety is 2.2 (Fig. 26). Each quill shaft has a shear neck, a deliberately thinned section whose lower breaking torque is at least 3.6 times the maximum rated torque, so that a seized accessory breaks its neck before any gear is harmed. The Phase 1 design needed three changes: the generator gear became a solid 16 mm blank because its thin web vibrated at the mesh frequency, its shaft bearing moved from the 6205 to the 6305 size (56,600 h), and the overspeed-governor pair was changed to 21/30 teeth at module 2 to leave the main resonance range. The starter-generator stays on the core's own gearbox.

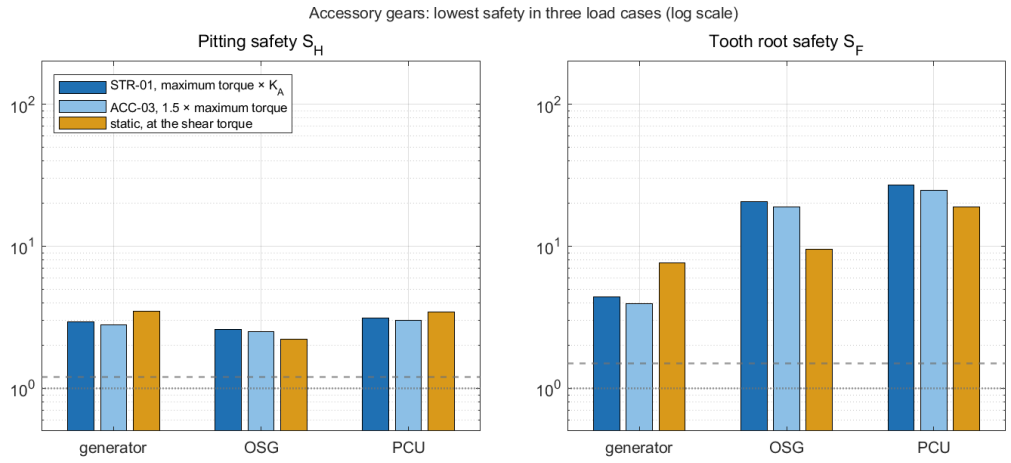


Fig. 26 Lowest pitting (left) and root (right) safety factors of the accessory gear pairs in three load cases (log scale).

G. Lubrication System

Oil does two jobs in a gearbox: it keeps the metal surfaces apart with a film thinner than a hair, and it carries away the heat of the losses. At full power the losses come to about 12 kW, the heat of six electric kettles boiling without pause, released inside a box about half a metre across.

A loss model with the mesh-loss formulation of Fernandes et al. [35], bearing friction after Palmgren and windage gives 12.4 kW at full power, an efficiency of 98.71%, and 14.9 kW of heat into the oil including the power turbine's

rear sump (Fig. 27, left); superfinishing lowered the mesh losses by about 19%. Thirty-two jets and feed holes of 0.6–1.6 mm need 23.5 L/min; the supply pump gives 30.5 L/min and the pumps absorb 2.4 kW. An air-oil cooler of 339 × 339 × 26 mm (4.4 kg) rejects 18 kW on a 53 °C day and returns the oil at 100 °C, the temperature on which the planet-bearing life depends (Fig. 27, right). On the ground an electric fan of about 0.5 kW supplies the cooling air, ram air in flight.

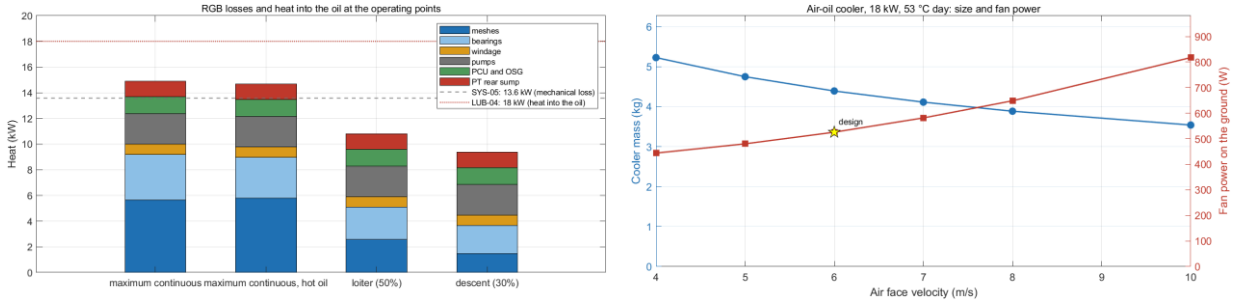


Fig. 27 Left: gearbox losses and heat into the oil at four operating points, by source. Right: cooler size and fan power against the air face velocity for 18 kW on a 53 °C day.

In Phase 3 the oil hardware was placed in CATIA and the hydraulic network was solved again on the real routes (Fig. 28). At full power all 32 jets receive 536–557 kPa; at ground idle the manifold still holds 296 kPa (Fig. 29, left). Two findings changed the design. After a start at −40 °C the jets pass no oil until it warms to about 11 °C, about 9 minutes at idle (Fig. 29, right), so the engine needs a minimum oil temperature before take-off power; and the oil tank must sit within 150 mm of the pump. For the requirement to keep running after a total loss of oil pressure, an emergency air-oil mist system (2.78 kg) gives 6.3–6.6 minutes at 75% power by analysis.

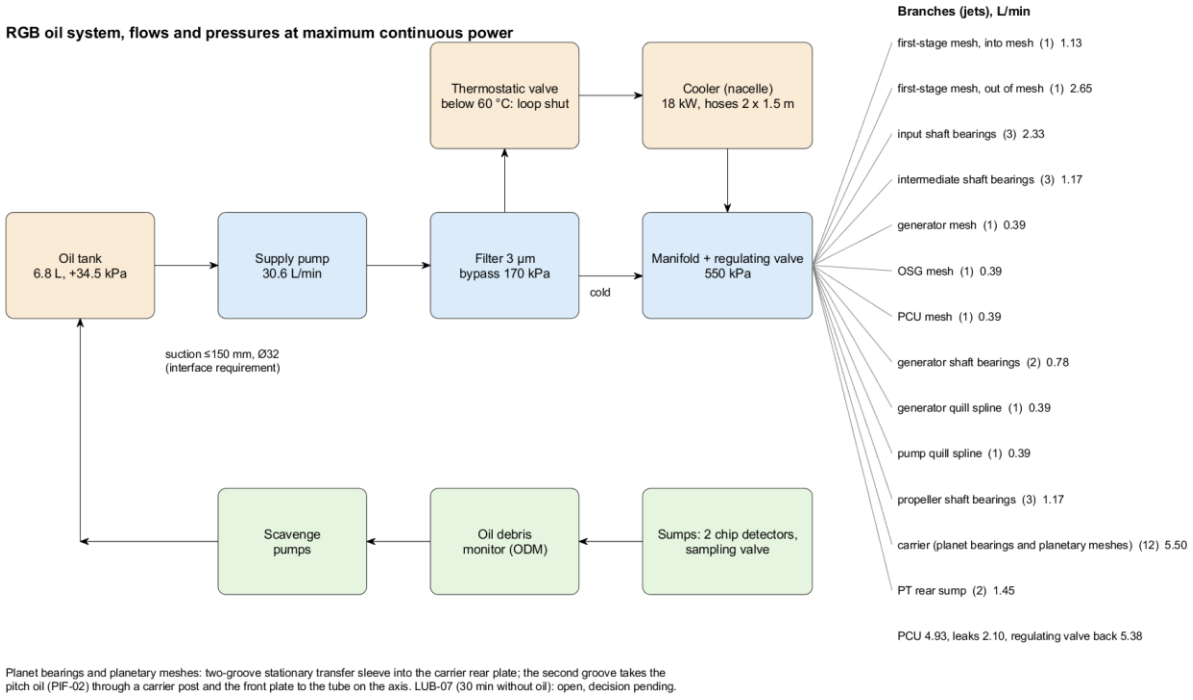


Fig. 28 Oil system with the flows and pressures at maximum continuous power: tank, supply pump, filter, cooler, thermostatic and regulating valves, manifold, jets, scavenge pumps and chip detectors.

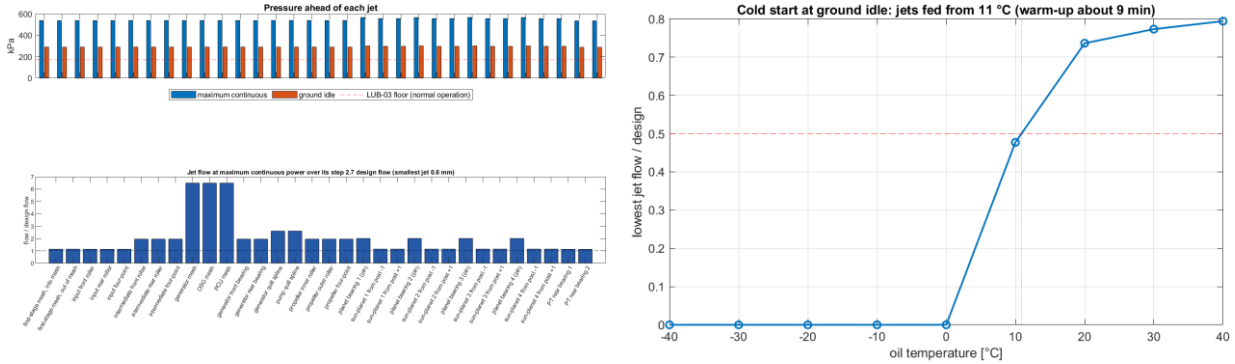


Fig. 29 Left: pressure ahead of each jet at full power and at ground idle, and the flow of each jet against its design flow. Right: cold start at ground idle; the jets are fed only from about 11 °C oil.

H. Cases

The case is the skeleton of the gearbox: it holds every bearing in its place. If it bends too much under load, the shafts tilt, the teeth no longer meet squarely and the propeller bearings lose the stiffness that keeps the shaft from whirling. Designing a case is therefore as much about stiffness as about strength.

The cases are A357-T6 aluminium sand castings; WE43 magnesium was 3.4 kg lighter but less stiff and harder to protect against corrosion and fire. Finite element models in the MATLAB PDE Toolbox [36] showed that the first case section (Fig. 30, left), with a flat front wall and a cylindrical nose, was too soft at the outer propeller bearing (3.3×10^7 N/m against the 5×10^7 N/m floor) and too highly stressed at its fillets (226–252 MPa against 165 MPa allowed). The case was redesigned with an 8 mm conical nose that runs down to the outer bearing boss (Fig. 30, right): it weighs 10.94 kg and gives 9.47×10^7 N/m at the outer seat, 6.56×10^7 N/m even with an opened bolted joint (Fig. 31).

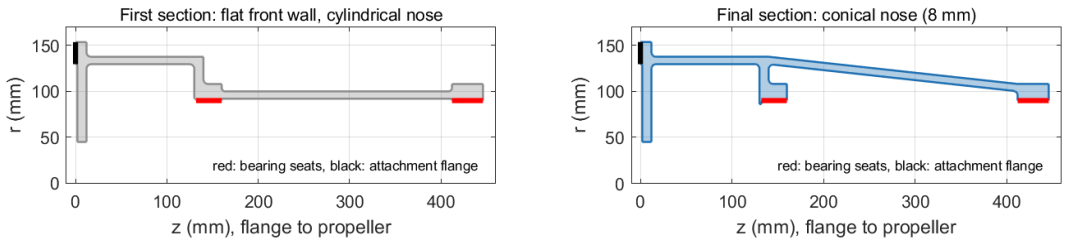


Fig. 30 Case sections: the first design with a flat front wall and a cylindrical nose (left) and the final conical nose (right).

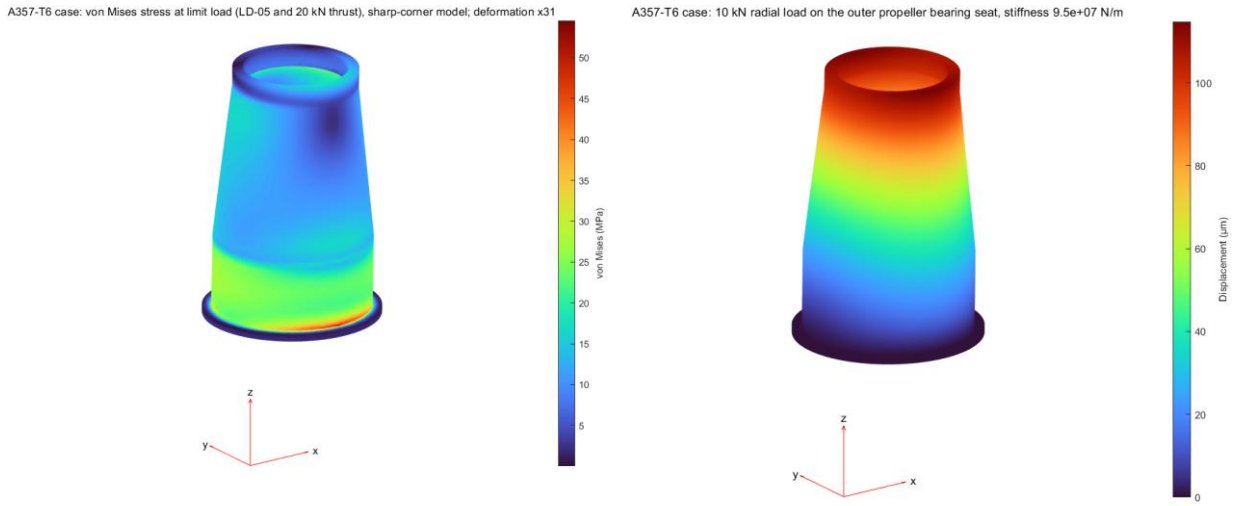


Fig. 31 The conical case under limit loads: von Mises stress (left) and displacement under a 10 kN radial load on the outer propeller bearing seat (right).

How far can such results be trusted? Two checks were made (Fig. 32). A plate with a hole, whose stress concentration is known exactly, was solved with the same tools: the computed factor converges to the theoretical 3.14 as the mesh is refined. And the most loaded corner, the fillet at the flange root, was solved again in a finely meshed submodel: 81 MPa at limit load, 128 MPa with a spring-supported flange, both below the allowable.

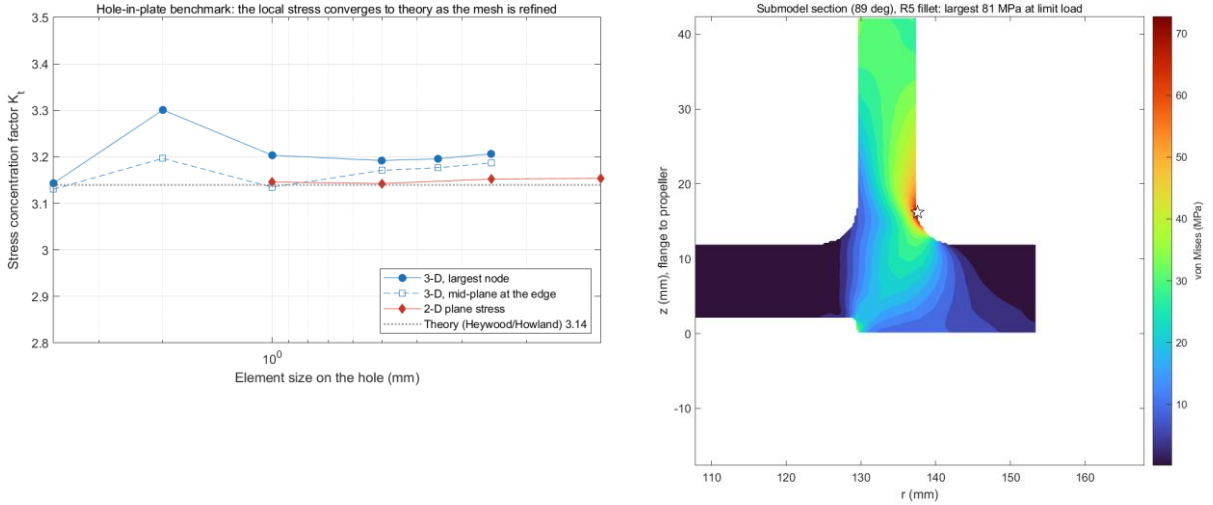


Fig. 32 Left: hole-in-plate benchmark; the computed stress concentration converges to theory. Right: submodel of the R5 fillet at the flange root at limit load (largest stress 81 MPa).

I. The Complete CAD Model and Its Mass

Up to this point every part had been designed on its own. Only when all of them come together in one model does it show whether they fit, and what the whole really weighs.

All parts were built in CATIA V5 by scripts that read the values exported from the MATLAB tools, so no dimension is typed by hand and a part is rebuilt in minutes when a calculation changes. In Phase 3 the model was completed with the first-stage case, the accessory cover, the generator cartridge, the bought parts and the oil system: 57 instances in three modules (Fig. 33), whose principal parts are shown one by one in Fig. 34. A clash analysis found no interference; all contacts are intended (bearing seats, meshes, splines, bolted faces).

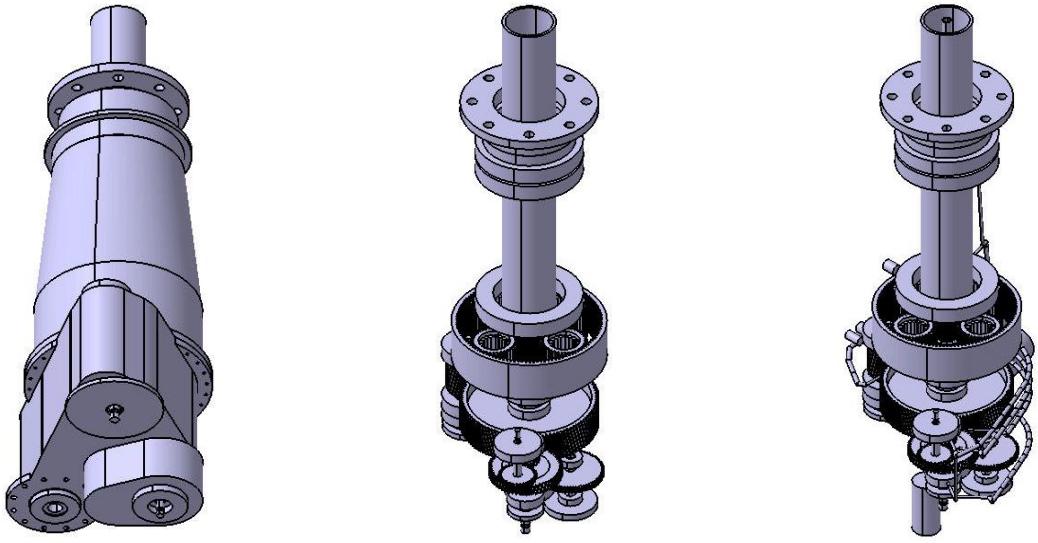


Fig. 33 The complete RGB in CATIA: planetary case and nose (left), and the gearbox with the accessory units seen from the engine side (centre and right).

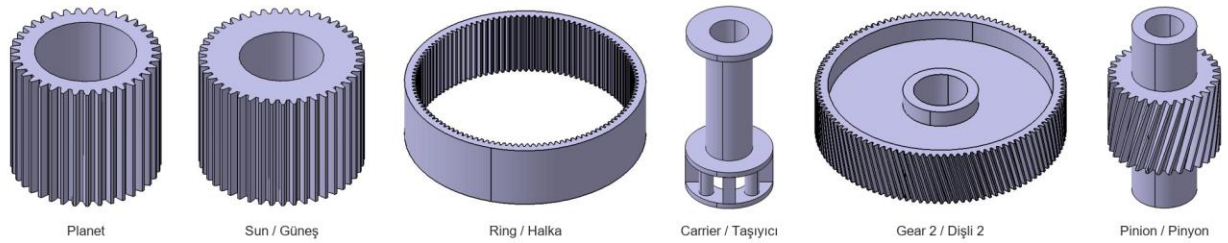


Fig. 34 Principal parts generated from the calculation parameters: planet, sun, ring, carrier with propeller shaft, first-stage gear, pinion, generator gear and case.

The first CAD weighing gave 114.6 kg against the original 100 kg limit; the "about 90 kg" of Phase 2 had been an estimate, not a weighing. It had been found by adding the Phase 2 growth to the PDR estimate of 77.6 kg (+3.0 kg for the propeller shaft, +3.5 kg for the wider planetary stage and its bearings, +3.6 kg for the retention tube, +1.5 kg for the accessory drives and +1.1 kg for the conical case). The full model showed what the estimate had missed: the first-stage case and the accessory drives, never modelled in Phase 2, fifteen bearings, the bolts, the oil hardware and the sensors. Mass almost always grows at the first complete CAD roll-up, which is why programmes trust weighings rather than estimates; Fig. 37 traces the whole history. A first lightening package (A: a lighter placeholder flange, thinner walls on the first-stage case and hollow planet pins) brought the model to 107.7 kg. Fig. 35 shows the further options that were considered: a titanium section of the propeller shaft (B), WE43 magnesium castings (C), or both (D, 95.2 kg). The author chose A: keep the materials and raise the limit to 110 kg (TBR), because B would cut the shaft bending stiffness by 44% and C would undo the case material decision. The emergency mist system and the weighed oil hardware then raised the roll-up to 111.1 kg (Fig. 36 shows where that mass sits); a second package (carrier plates thinned from 15 to 10 mm and a bottom-up count of the fasteners instead of a flat allowance) and smaller later changes brought it to the final 109.95 kg. The 5.46 kg drive shaft between the core and the gearbox is booked separately. The 100 kg limit itself had come from a rough Phase 0 scaling (the CT7-9 is about 167 kg heavier than the T700-GE-700 turboshaft; scaled to the lower torque, 93–105 kg), so the 110 kg limit stays TBR until it is agreed with the customer, and any further rise, at most to 115 kg, needs a new decision by the author.

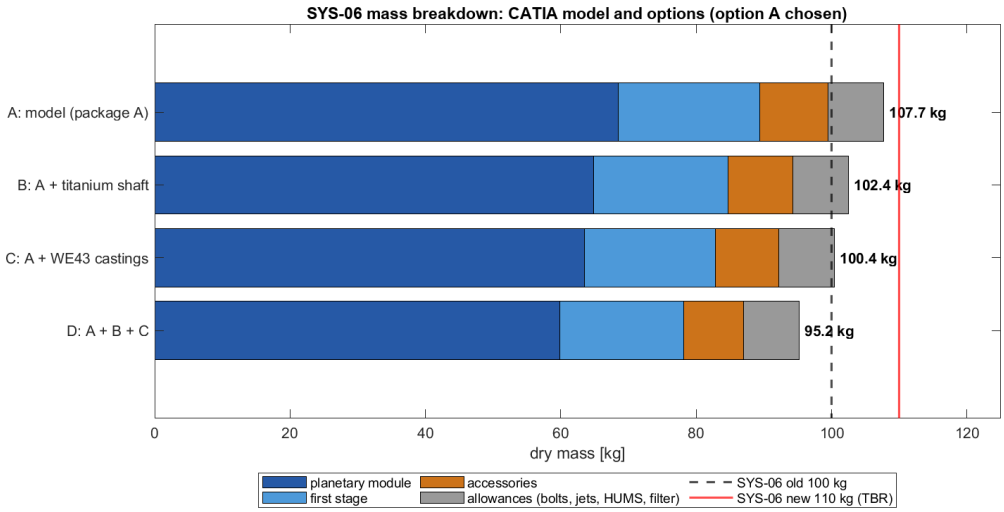


Fig. 35 Mass of the CAD model after package A and of the three further options, against the original 100 kg and the new 110 kg (TBR) limit; option A was chosen.

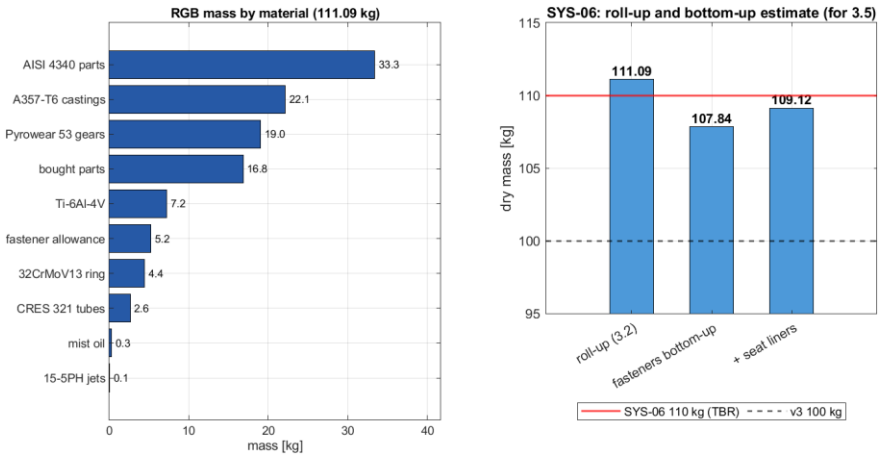


Fig. 36 Left: gearbox mass by material at the roll-up of step 3.4 (111.09 kg); AISI 4340 parts, mainly the carrier with the propeller shaft, and the A357 castings dominate. Right: that roll-up against the bottom-up fastener estimate with the seat liners, the starting point of the second mass package.

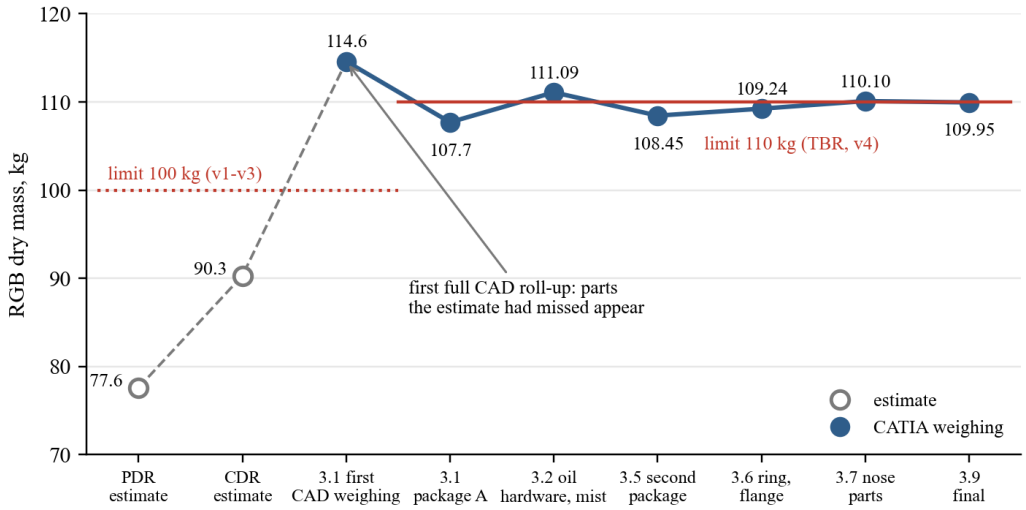


Fig. 37 How the gearbox mass evolved through the project: estimates (open markers) and CATIA weighings (filled markers), with the 100 kg limit of requirements v1–v3 and the 110 kg (TBR) limit set in step 3.1.

VI. Structural and Dynamic Verification by Analysis

With the complete model available, Phase 3 checked the main load-carrying parts in three dimensions and closed the vibration requirements with a forced-response analysis. In short: the planetary case and the carrier have wide static margins; the analysis moved the engine mounts and added dowels to the case flange; and the vibration of the gear web and the case is acceptable after one design change and a friction damping ring.

A. Three-Dimensional Structural Analysis

Hand formulas work for simple shapes. A cast case with ribs, bosses and holes needs the finite element method: the part is cut into a great many small tetrahedra, the deformation of each is solved together with its neighbours, and the stresses follow.

The planetary case, the first-stage case and the carrier with the propeller shaft were meshed from their CATIA geometry with quadratic tetrahedra and loaded with the limit cases of Section III. Fig. 38 summarises the safety factors of every check. The planetary case shows a limit stress of 36 MPa (yield safety 5.4 including a casting factor of 1.25), an outer seat stiffness of 1.0×10^{8} N/m and a once-per-revolution fatigue safety of 5.2. The carrier keeps a yield safety of 3.1 in the sudden-stoppage case with its planet pins pressed in.

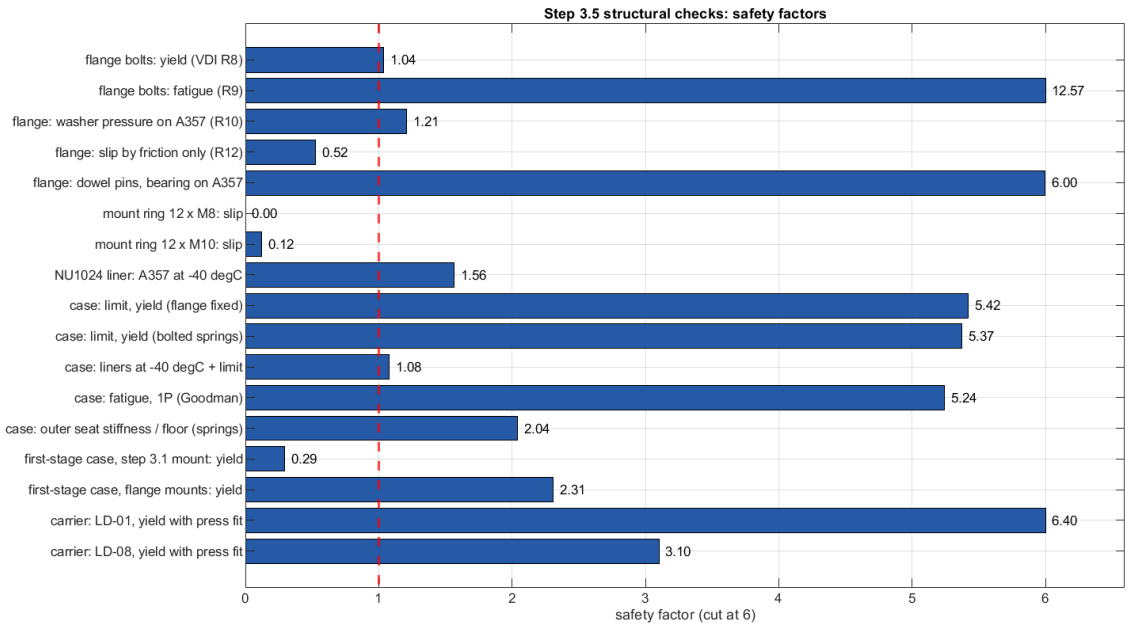


Fig. 38 Safety factors of all 3-D structural checks of Phase 3 (dashed line = 1). The lowest bars, the old engine-mount ring and the flange friction, led to design changes.

The analysis found five actions. The most important concerned the engine mounts: in the first layout they were on a ring around the power-turbine axis, so the whole propeller load passed through the 6 mm walls of the first-stage case, giving 653 MPa and a yield safety of only 0.29 (Fig. 39, left). The author chose to move the mounts to the planetary-case flange and to thicken the first-stage rear wall to 16 mm, after which that case sees 29 MPa. The 30 M6 bolts of the case flange could not carry the limit torque by friction alone (slip safety 0.52 after VDI 2230 [37]), so the flange received six Ø10 dowel pins. A steel bearing liner needed a 3.8 mm thicker aluminium boss for its fit at −40 °C, and a Phase 2 load case was corrected in which the two gyroscopic seat reactions had been applied in the same direction.

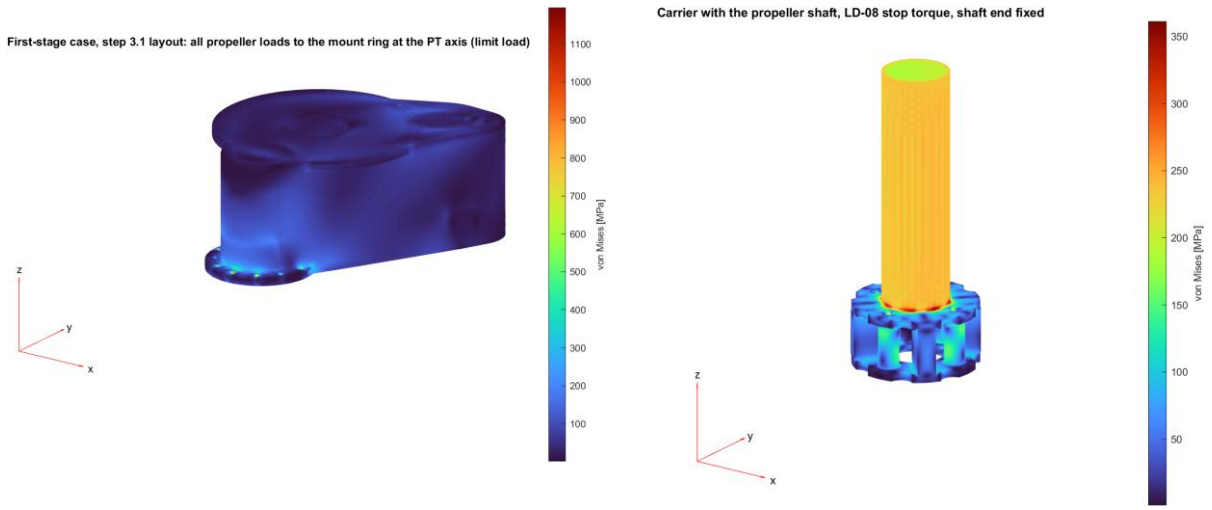


Fig. 39 Left: first-stage case with the propeller loads taken at the old mount ring on the power-turbine axis (limit load); the stress led to moving the mounts. Right: carrier with the propeller shaft under the sudden-stoppage torque, shaft end fixed.

The carrier was lightened in two steps to save mass. First, six scallops were cut into its plates: the deflection at the planet pins rises only from 29.8 to 30.8 µm (Fig. 40). Later the plates were thinned from 15 to 10 mm, the author's choice among three mass packages: the pin tilt over the 70 mm face grows from 19.1 to 25.8 µm and the pitting safety

of the sun-planet mesh falls from 1.34 to 1.28, still above 1.2. The sudden-stoppage yield safety away from one unfilleted corner is 2.53; the drawing specifies a fillet of at least R5 there.

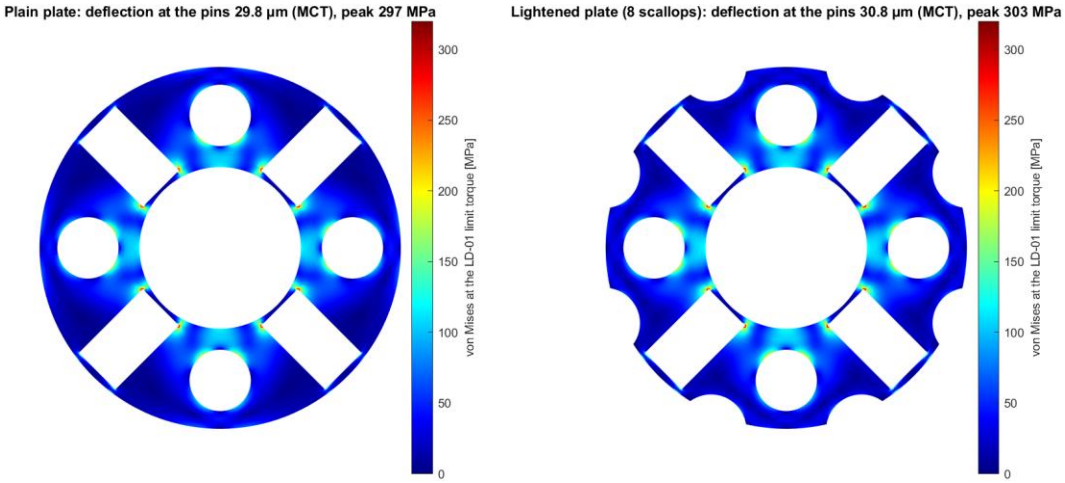


Fig. 40 Carrier front plate at maximum continuous torque, plain (left) and with six scallops (right): deflection at the pins 29.8 and 30.8 µm.

B. Forced Response to the Transmission Error

Even when nothing is wrong, a gearbox hums. Every tooth contact gives the structure a tiny knock, thousands of times a second; if a part has a natural frequency near that knocking rate, it rings like a bell, and ringing means alternating stress.

> **Term: transmission error**
> Even perfect-looking gears do not transmit motion perfectly smoothly: under load the teeth bend, and their profiles are never exact. The output gear therefore runs slightly ahead or behind its ideal position, a few tenths of a micrometre, once per tooth contact. This transmission error (TE) is the main source of gear vibration; it repeats at the mesh frequency and its harmonics.

The excitation was modelled as the TE of each mesh at its first three harmonics: 0.5 / 0.15 / 0.1 µm for the planetary meshes and 0.5 / 0.2 / 0.1 µm for the first stage, plus the axial shuttle of the helical line of action (all TBR). The second planetary harmonic became a new requirement, STR-11, to be shown by a loaded tooth contact analysis and a loaded TE test. A three-dimensional planet model with exact spring tensors for its rollers and meshes gave planet modes at 3,309, 4,008, 6,658 and 7,370 Hz, correcting the Phase 2 values. Every mesh band holds a planet mode (Fig. 41, left), so frequency separation cannot be met and acceptance rests on the dynamic tooth force (Fig. 42, left): the governing dynamic factors are 1.11 / 1.37 / 1.14 at 1× / 2× / 3×, and the planet tooth-root safety stays at or above 1.55.

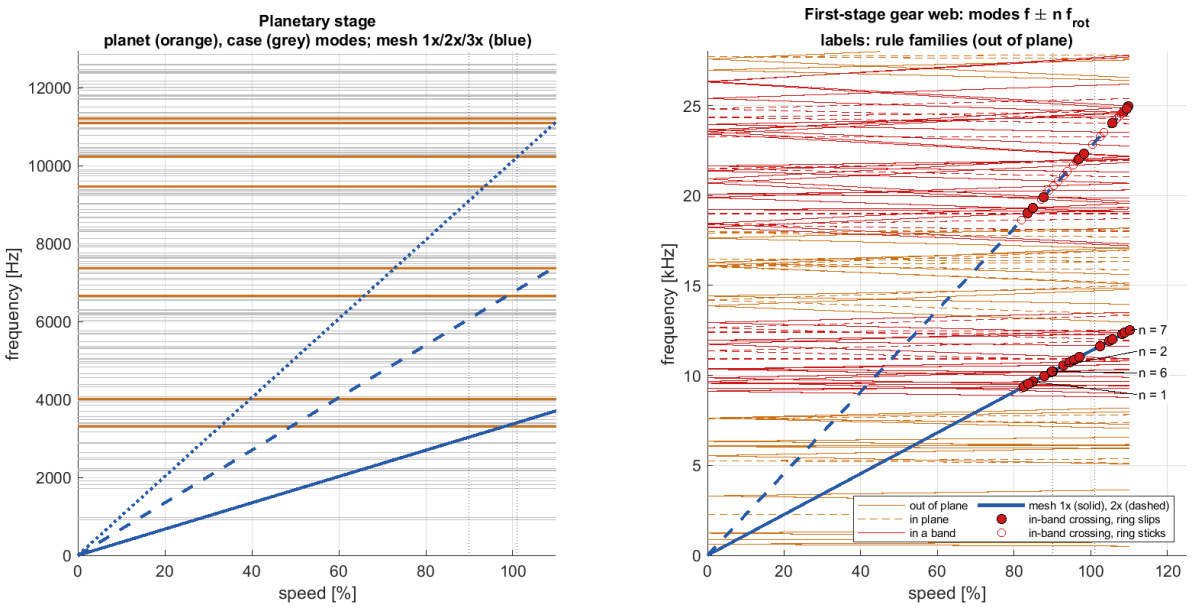


Fig. 41 Left: planetary stage, planet (orange) and case (grey) modes against the mesh lines (1×, 2×, 3×). Right: out-of-plane modes of the first-stage gear web; the labelled families slip under the damping ring.

Low memory on the analysis computer led to axisymmetric harmonic finite element models for the planetary case and the first-stage gear web, a method that represents a ring-like part by its cross-section and a number of waves around the circumference. The case was coupled to the four planets through the ring-seat receptance with the mesh phasing of the 119-tooth ring; the static planet loads that rotate with the carrier add 12.2 MPa of alternating stress to a steady 16.7 MPa. Node by node, the Goodman factor of the A357 shell is at least 1.19 at the flange junction (Fig. 42, right), after one design change: the flange bolt circle grew from 290 to 308 mm (outer diameter 325 mm) so that an R10 outer and R5 inner junction fillet could be built.

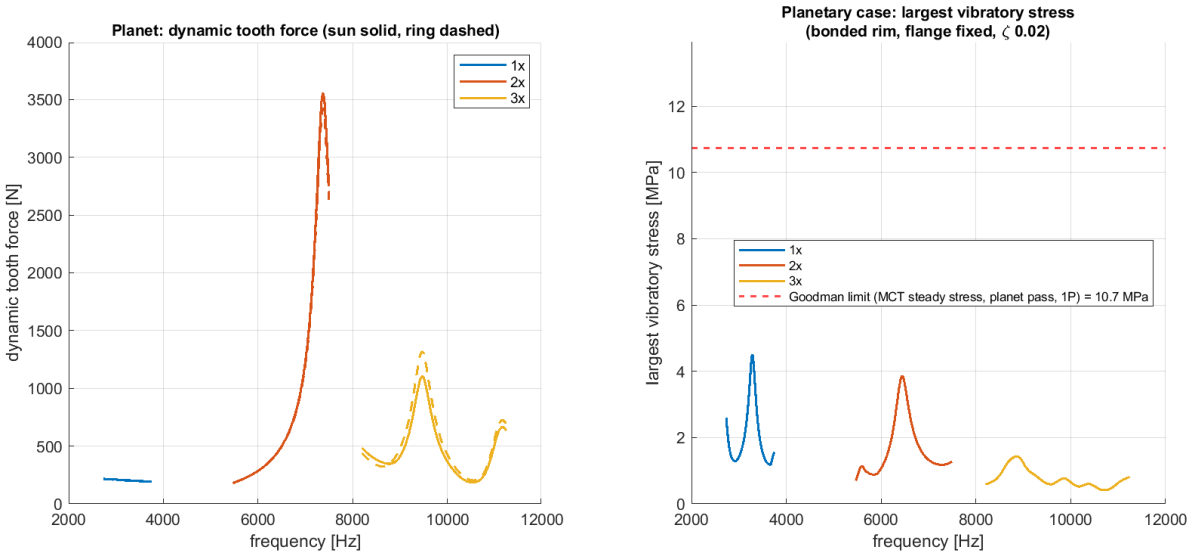


Fig. 42 Left: dynamic tooth force of the planet against the excitation frequency (sun side solid, ring side dashed). Right: largest vibratory stress in the planetary case with the ring bonded and the flange fixed.

The first-stage gear web has in-band modes that cannot be avoided. A split friction damping ring, pressed into a groove by its own centrifugal load, dissipates energy by slipping when the web vibrates. It was sized with a slip model that includes a stick limit and pre-sliding: the lightest ring that keeps the amplitude ratio at or below 0.6 for the slipping mode families is 8 × 4 mm (0.164 kg), and it lowers the vibratory stress of those families substantially (Fig. 43); the web Goodman factor rises from 1.96 bare to 2.31.

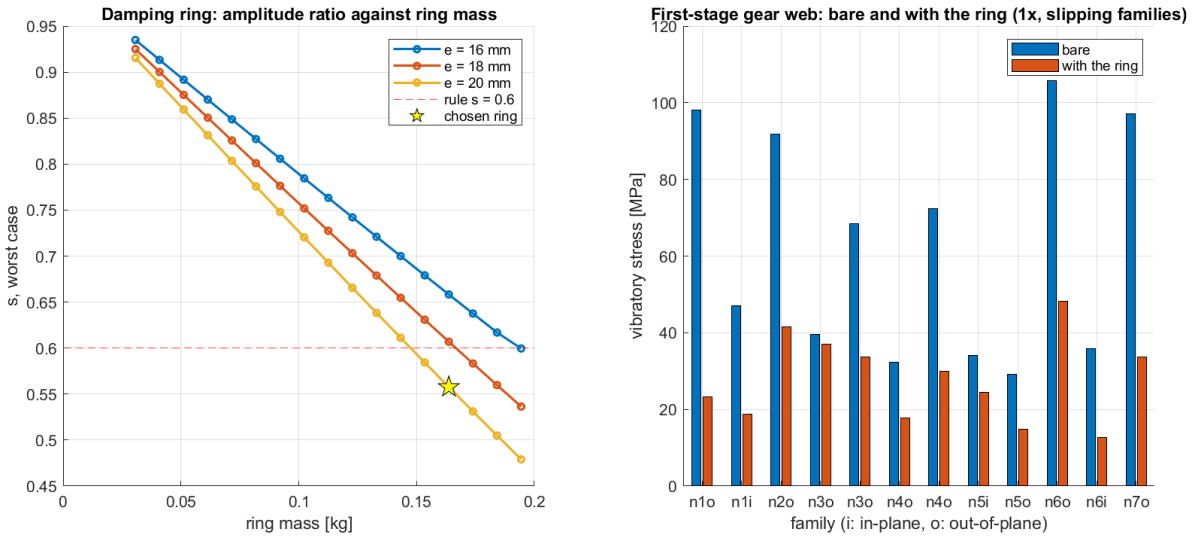


Fig. 43 Left: amplitude ratio of the slipping mode families against the ring mass for three ring sizes; the chosen 8 × 4 mm ring is marked. Right: vibratory stress of each web mode family, bare and with the ring.

VII. Safety and Certification Analyses

An engine must show that its failures cannot endanger the aircraft except at extremely low rates. Four analyses address this: a fault tree for the worst event, an FMEA for every failure mode, a critical-part and damage-tolerance assessment, and a fire assessment.

A. Fault Tree for Propeller Release

The worst thing this gearbox could do is let go of the propeller. Showing that this is practically impossible means tracing every route that could lead there.

> **Term: fault tree and cut set**
>
> A fault tree starts from an unwanted top event, here "the propeller separates from the engine", and asks which combinations of failures can cause it. A cut set is a smallest such combination. AC 33.75-1A [38] requires every cut set to stay at or below 10^{-8} per engine flight hour and the sum of all hazardous effects at or below 10^{-7}; failures that stay hidden until an inspection count with their average unavailability over the inspection interval.

The propeller is held by three paths: the primary path (propeller shaft, four-point contact bearing and its nut), a standby axial path (a roller-bearing rib and a titanium catcher lip in the nose cover, normally unloaded) and the titanium retention tube. A thrust floor in the engine control keeps the standby path unloaded in service, and a limp mode caps the thrust after a bearing thermocouple alert (CTL-04). The largest quantified cut set is 2.1×10^{-9} per flight hour and the sum 3.3×10^{-9}, inside the extremely remote band (Fig. 44). Separate trees for power-turbine overspeed and for loss of the propeller overspeed protection led to rewritten requirements: two independent protection speed signals from the core with 1-out-of-2 trip logic (SAF-03), and four propeller protection layers with unavailability budgets for the propeller maker (PIF-05).

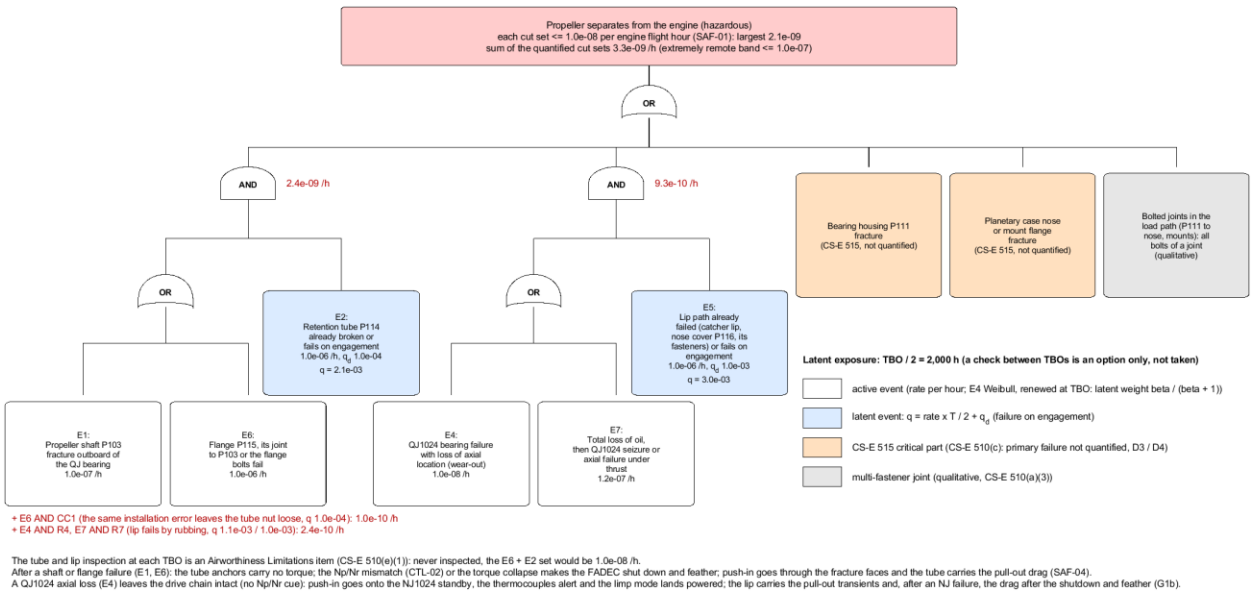


Fig. 44 Fault tree of propeller separation with the cut-set probabilities per engine flight hour.

B. Failure Mode and Effects Analysis

Where the fault tree starts from the worst event and works downward, the FMEA works upward: it takes every part in turn and asks how it can fail, what happens then and how the failure would be noticed.

The FMEA, in the style of ARP5580 [39] but with the CS-E 510 effect classes instead of risk priority numbers, lists 100 failure modes in ten groups: 13 hazardous, 44 major and 43 minor after the design features; without them 51 modes would be hazardous (Fig. 45, left). Every hazardous mode is covered by a critical part or a fault-tree cut set. Fig. 45 (right) shows how the failures would be noticed: vibration monitoring, the oil debris monitor, oil pressure and the chip detectors name most of them, and every latent mode has an inspection task in the maintenance instructions.

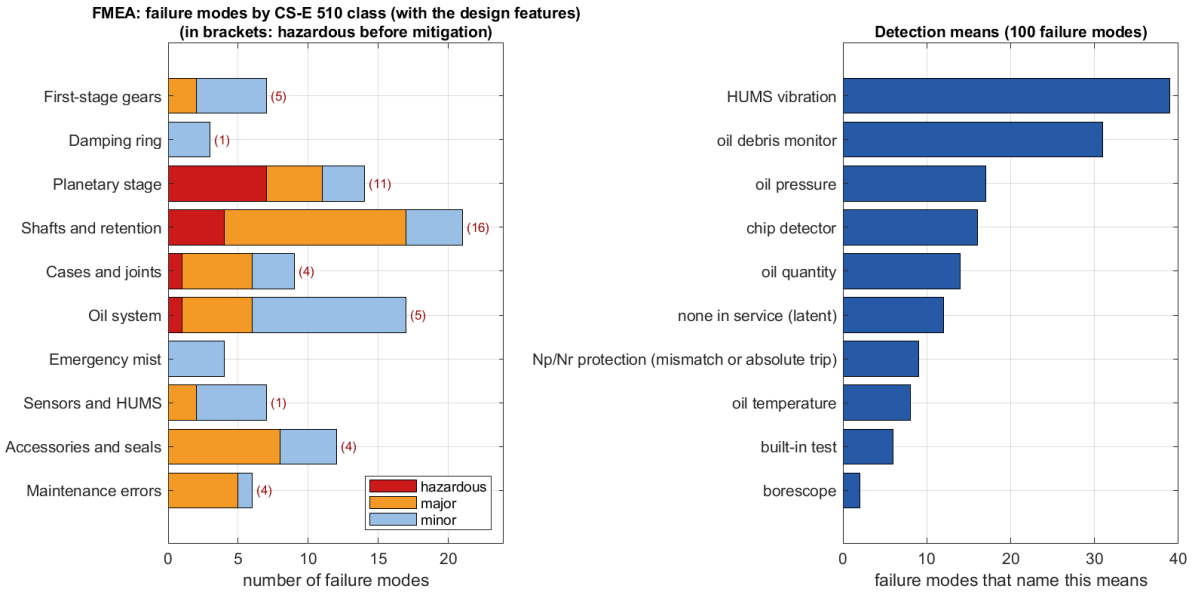


Fig. 45 Left: failure modes per group by CS-E 510 class with the design features (in brackets: hazardous before mitigation). Right: how many failure modes each detection means can reveal.

C. Critical Parts and Damage Tolerance

Damage tolerance does not assume a perfect part. It assumes that a small crack is already there on the first day, too small for inspection to find, and asks how long it would take to grow to a dangerous size. If that time is long compared with the inspection interval, the crack is found before it matters.

Seven parts are critical under CS-E 515: the power-turbine output shaft, the gearbox input shaft, the carrier with the propeller shaft, the propeller bearing housing, the planetary case, the shoulder ring and the two diaphragm couplings. Each has a damage-tolerance location, a non-destructive inspection plan at manufacture and at overhaul, and a life basis. Following CM-PIFS-007 [40], a small surface flaw of 0.762 × 0.381 mm was assumed at each location and grown from the first cycle with the Newman–Raju stress-intensity solutions [41] (Fig. 46). The shortest life, at the coupling hub, is 7,195 cycles against a target of 3,000. At three locations the crack that would just start growing under vibration is smaller than inspection can reliably find, so these rely on process control (hot isostatic pressing, CT or radiography, a 0.5 mm pore limit) instead.

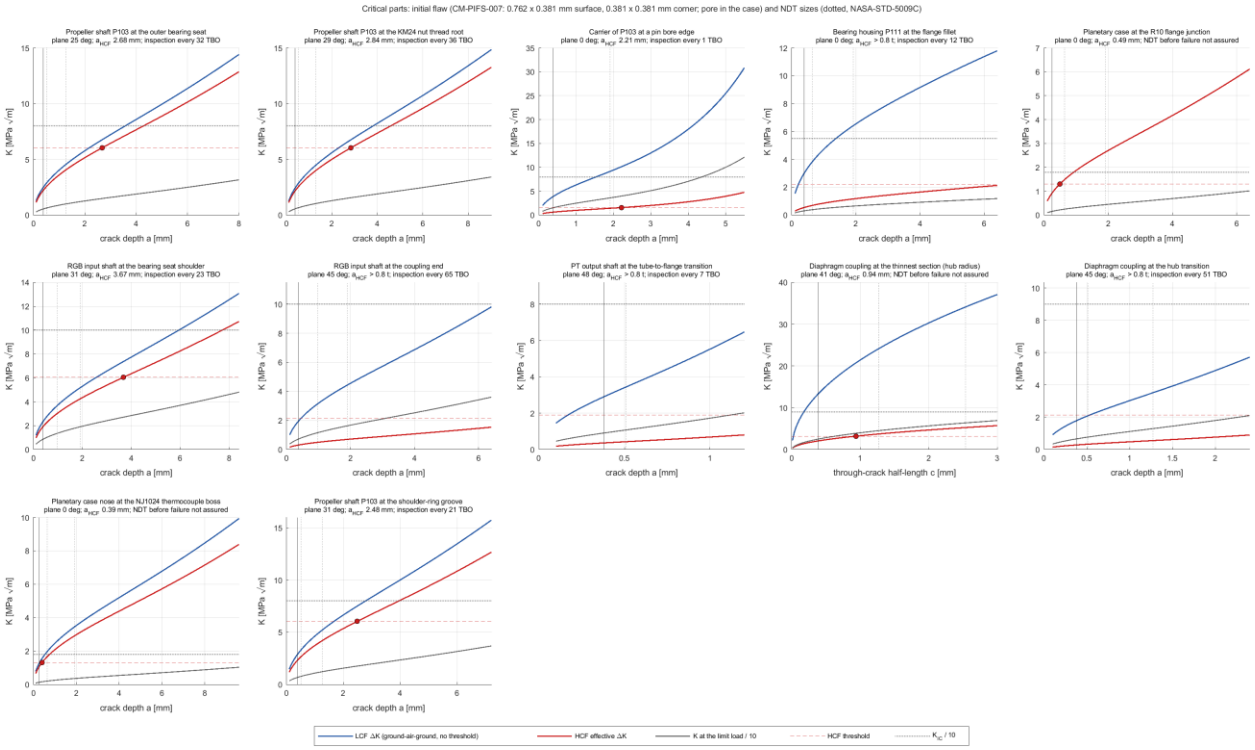


Fig. 46 Crack driving force against crack depth at the twelve critical locations, compared with the material threshold and toughness.

D. Fire

An oil-filled gearbox next to a hot engine must survive a fire long enough for the engine to be shut down: the rule is five minutes in a standard 1,100 °C flame.

The first-stage side of the gearbox lies in a designated fire zone; the planetary case stays outside it behind a partition at the case joint. In the ISO 2685 standard flame [42] (1,100 °C, 116 kW/m^2) the oil-wetted A357 walls survive five minutes while the oil flows and cools them (Fig. 47). Walls without an oil film reach the melting range of A357 in 56 s, the joint O-rings reach about 360 °C against a 204 °C rating, and after the oil flow stops the walls reach the melting range in 24–31 s. Wall-spray jets, a fire-rated secondary seal and a fire test are planned; the fire requirement SAF-05 remains open.

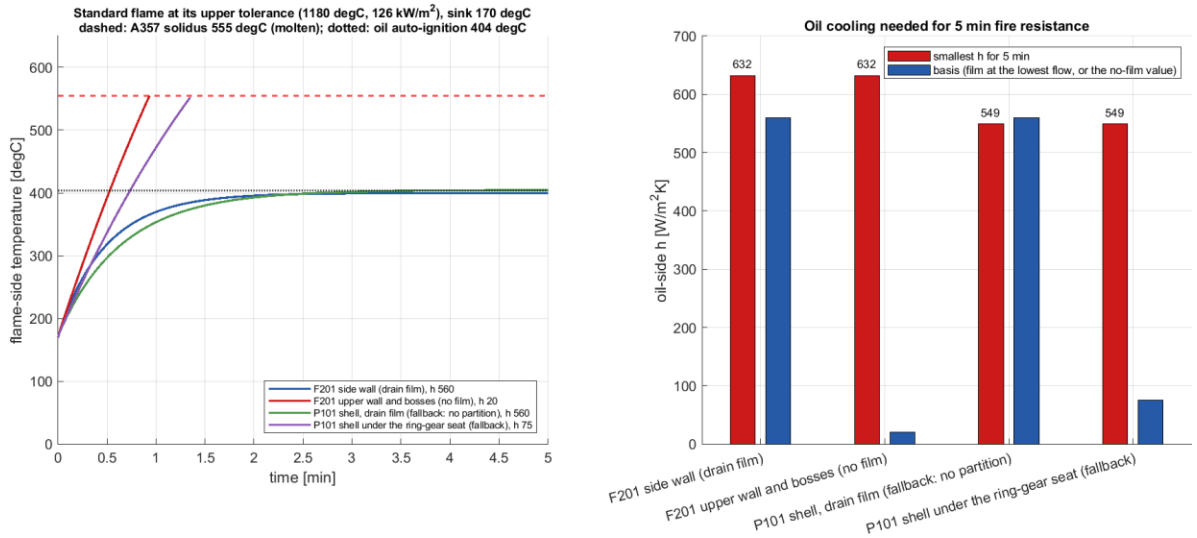


Fig. 47 Left: wall temperatures in the standard flame with and without oil cooling. Right: oil cooling needed by each wall for five minutes of fire resistance.

VIII. Manufacturing Definition

A design is only finished when it can be made and inspected. Phase 3 gave every part a material, a process route, tolerances and an inspection, and produced representative drawings.

A. Materials, Processes and Bill of Materials

A gear tooth needs two opposite things: a surface hard enough to resist wear and contact fatigue, and a core tough enough not to snap under a shock. Carburizing resolves this by soaking carbon into the surface and hardening only that skin, much like the hard crust on soft bread.

Table 12 summarises the key choices; the full bill of materials has 78 lines. Carburizing gives a gear a hard, wear-resistant case on a tough core; the depth of that case must exceed the depth at which the contact stress peaks below the surface. Fig. 48 shows this check: the planet tooth on the ring side needs 0.43 mm at the over-torque load, so the specified case depth for the main gears is 0.5–0.7 mm. Because the planet rollers run directly on the planet pins, the pins are separate carburized M50NiL parts pressed into the AISI 4340 carrier, which cannot itself serve as a raceway.

Table 12 Principal materials and processes.

Part	Material	Process and inspection
Gears (pinion, gear, sun, planets, accessory gears)	Pyrowear 53, AMS 6308	carburized 58–63 HRC, case 0.5–0.7 mm, superfinished Ra ≤ 0.15 µm; temper etch, magnetic particle
Ring gear	32CrMoV13 (EN 1.8523)	gas nitrided, ≥ 650 HV, 0.5–0.7 mm; honed and superfinished
Carrier with propeller shaft	AISI 4340, AMS 6414	quenched and tempered to 1,240 MPa; ultrasonic and magnetic particle; critical part
Planet pins	M50NiL	carburized, pressed into the carrier
Retention tube	Ti-6Al-4V	second load path; inspected at each overhaul
Cases	A357-T6 sand casting	hot isostatic pressing, CT or radiography, 0.5 mm pore limit, steel seat liners

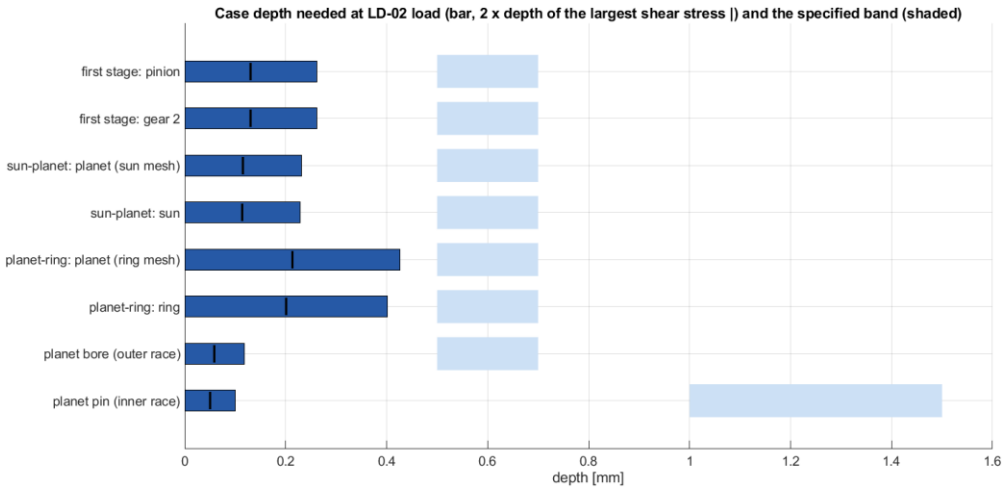


Fig. 48 Case depth needed at the over-torque load (bar: two times the depth of the largest shear stress) against the specified band (shaded).

B. Tolerances, Stack-Ups and Fits

No part is ever made exactly to its drawing; every dimension may vary within a small band. When many parts are stacked along a shaft, these small variations add up, and at −40 °C or with hot oil the aluminium case and the steel parts grow by different amounts. Tolerance analysis checks that, in every such combination, fits stay tight where they must and gaps stay open where they must.

Datums and geometric tolerances follow ASME Y14.5-2018 [43], fits ISO 286, gear accuracy ISO 1328-1:2013 [44] and splines ISO 4156. Because the cases are aluminium and most other parts steel, every fit changes with temperature; a lumped temperature model of five states, from a −40 °C cold soak to the over-torque case with 149 °C oil, drives the fit calculations (Fig. 49, left). Stack-ups, the sums of many small tolerances along a chain of parts, were evaluated both worst-case and by Monte Carlo with the tolerances as ±3σ. The backlash of the gears stays positive through the cold-start transient and the planet load-sharing factor with the drawing tolerances is 1.257 at P99 (Fig. 49, right); the ring-gear seat stays tight up to the over-torque case (Fig. 50). The chain also showed that the titanium catcher lip in the nose cover could touch its mating part when hot; the drawing step opened that clearance to 0.80–1.00 mm. The 72 resulting callouts feed the drawings.

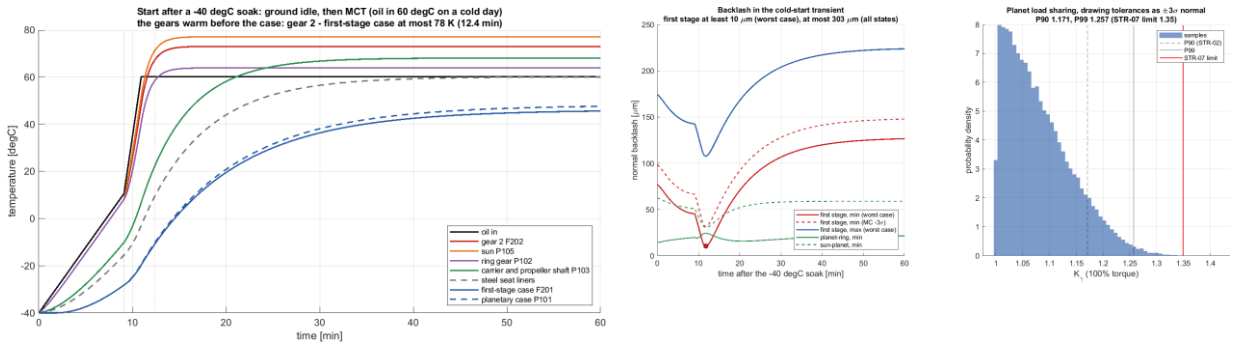


Fig. 49 Left: part temperatures after a start from a −40 °C cold soak. Right: gear backlash during the cold-start transient and the distribution of the planet load-sharing factor with the drawing tolerances.

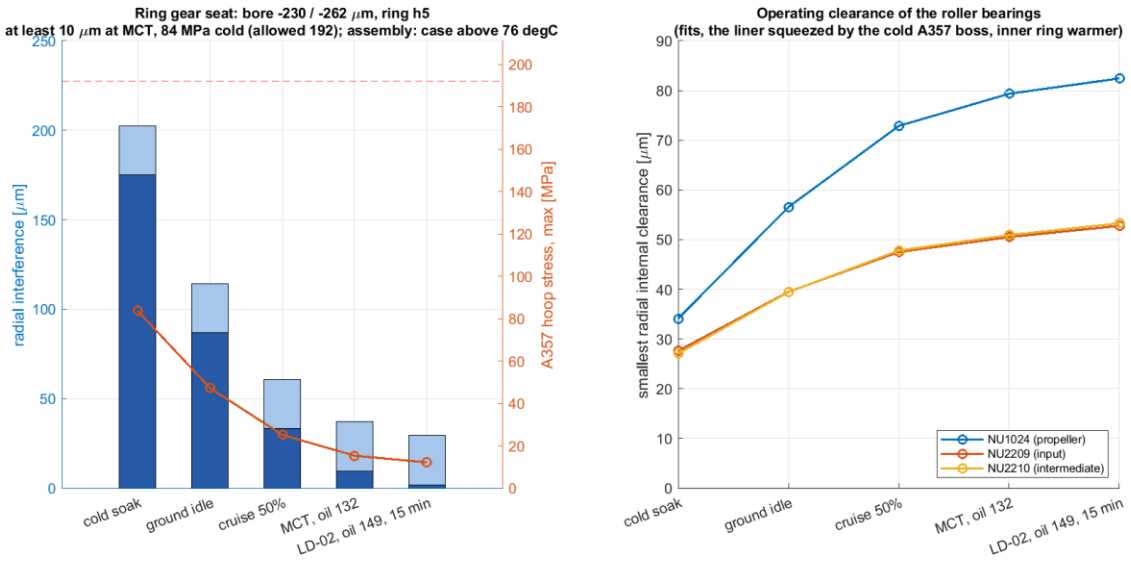


Fig. 50 Ring-gear seat interference and hoop stress (left) and roller-bearing operating clearance (right) over the five thermal states.

C. Drawings

The drawing is the contract with the workshop: it holds everything needed to make and inspect the part, and leaves nothing open to interpretation.

A representative set of four drawings was produced in CATIA with a bilingual title block: the planet gear, the carrier with the propeller shaft (a critical part, Fig. 51), the planetary case (casting and machining) and the gearbox assembly with its 89-line parts list (Fig. 52). Every value on the sheets traces to its source table through a 114-row checklist. The views were generated from the Phase 2 model; the Phase 3 geometry changes are given in the sheet tables, which govern.

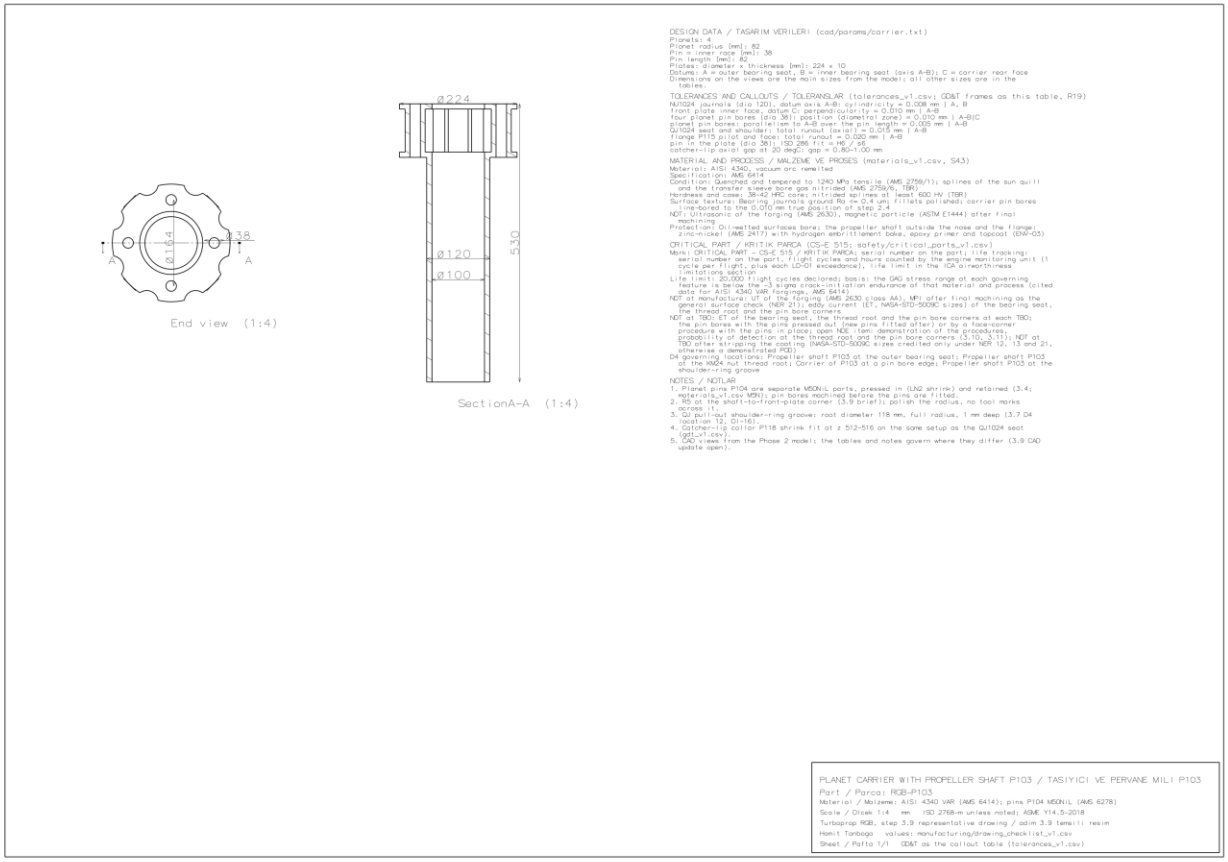


Fig. 51 Drawing of the carrier with the propeller shaft (critical part): views, design data, tolerances, material, NDT and critical-part notes.

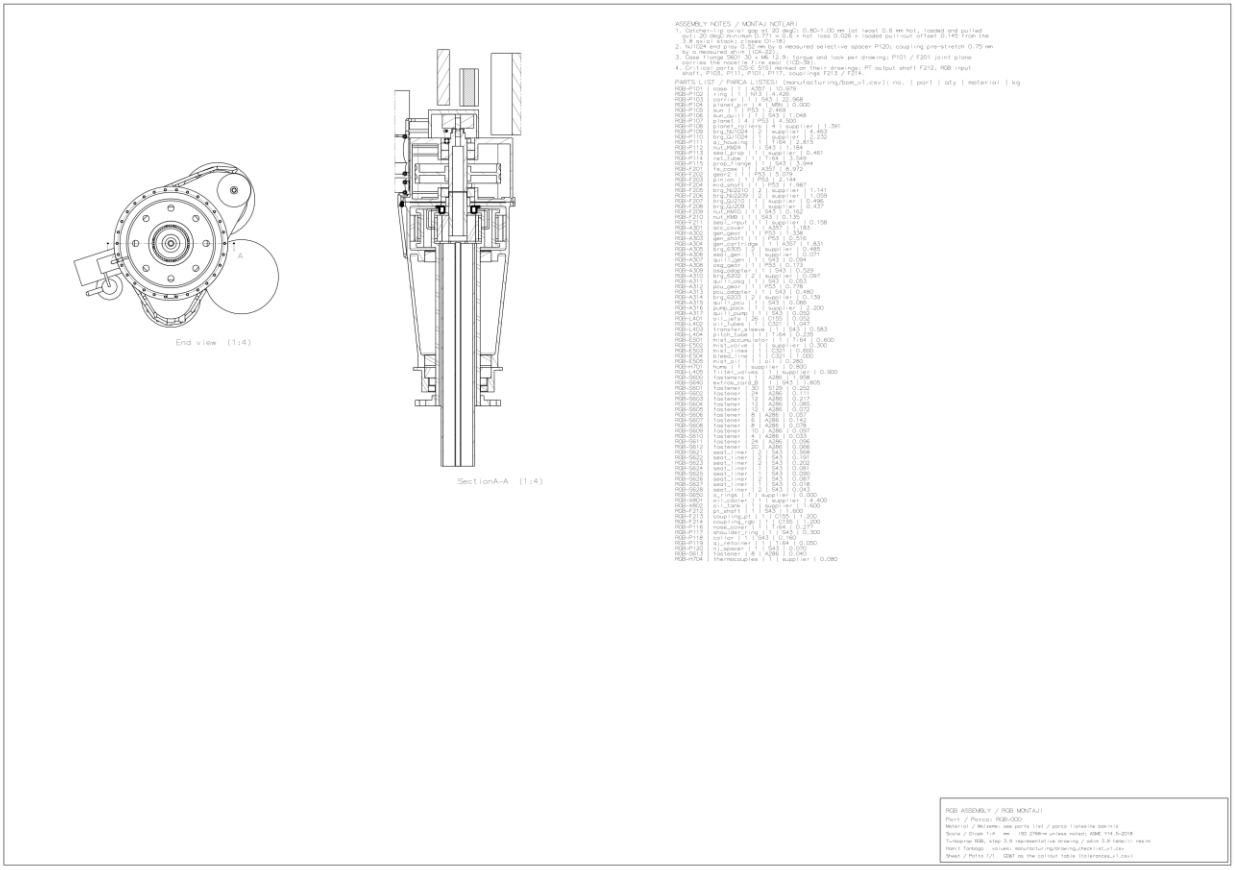


Fig. 52 Assembly drawing of the reduction gearbox with its parts list.

IX. Maintainability and Condition Monitoring

The customer flies long missions from forward bases, so the gearbox must be quick to change and should announce damage before it becomes dangerous. This was the innovation package of the project, chosen by the author.

A. Modular Maintenance and Access

In the field, a gearbox is judged not only by how long it lasts but by how quickly it can be changed when it must be.

The gearbox is a separate module of three sub-modules. A standard-time model built on the fastener, connector and line counts of the bill of materials gives 2.6 h to remove the gearbox from the aircraft and 4.7 h to install it with two technicians, including the coupling shim, an independent inspection, the oil fill and a ground run; the maintainability requirement was therefore set to 8 h (TBR). Because the engine mounts are on the gearbox flange, the core needs a support cradle while the gearbox is off. With the gearbox installed, two borescope ports, the chip detectors, the oil sample valve, the filter, the sensors and the accessories are all reachable. A ray-casting check on the CAD parts confirmed that the planetary port sees all 56 target points of the ring–planet mesh and the first-stage port 39 of 45 points of its mesh (Fig. 53).

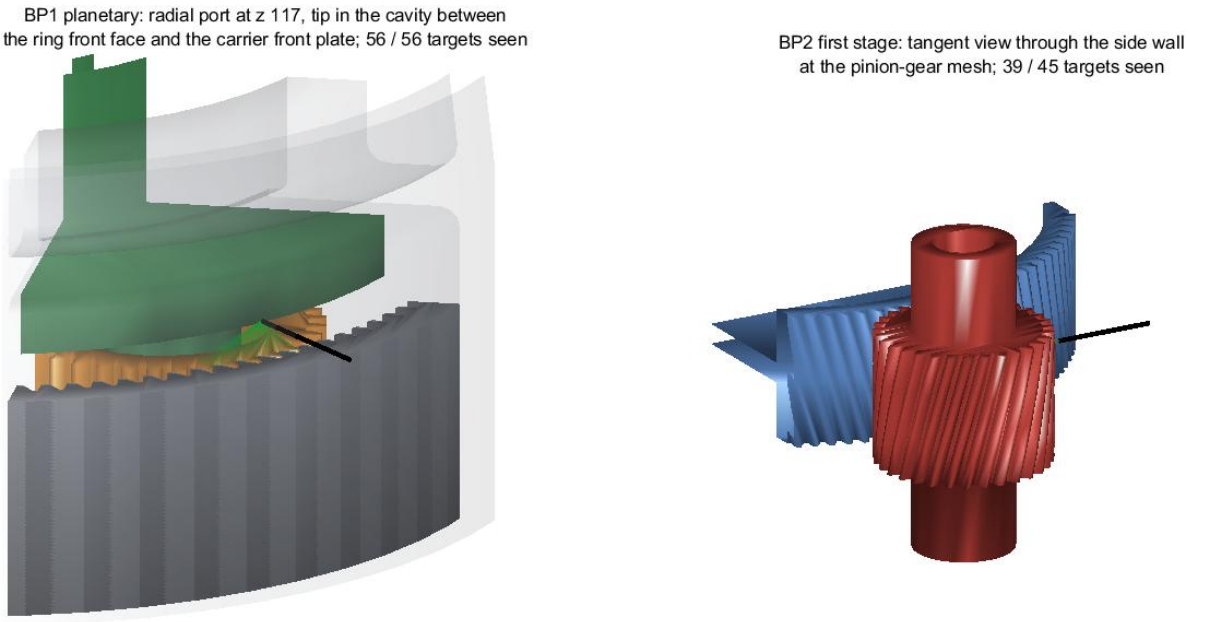


Fig. 53 Borescope lines of sight on the CAD parts: port BP1 into the gap in front of the ring gear (left) and port BP2 along the tangent of the first-stage mesh (right).

B. Where to Listen: Fault Frequencies and Sensor Positions

A gearbox sounds different when something inside begins to fail, just as a car's wheel bearing starts to hum before it gives way. Condition monitoring turns that idea into numbers.

> **Term: HUMS and condition indicators**
> A health and usage monitoring system (HUMS) listens to the gearbox with accelerometers and counts metal particles in the oil. A crack or a spall changes the vibration in a characteristic way: each shaft, mesh and bearing produces lines at its own frequencies. Condition indicators are numbers computed from the signal that stay steady on a healthy gearbox and grow when damage appears; an alarm is raised when an indicator crosses a threshold.

The characteristic frequencies of every shaft, mesh, local gear fault and the eleven bearings were computed (Fig. 54). Twenty-three pairs of lines lie within 2% of each other, so a simple spectrum cannot tell them apart; each shaft must instead be analysed by time-synchronous averaging (TSA), which averages the signal over many revolutions of that shaft and cancels everything not locked to it. The finite element model of the case gave the transfer from each vibration source to the case surface (Fig. 55), from which two accelerometer positions were chosen, on the case wall over the ring gear and on the outer propeller bearing boss, plus a third on the first-stage case.

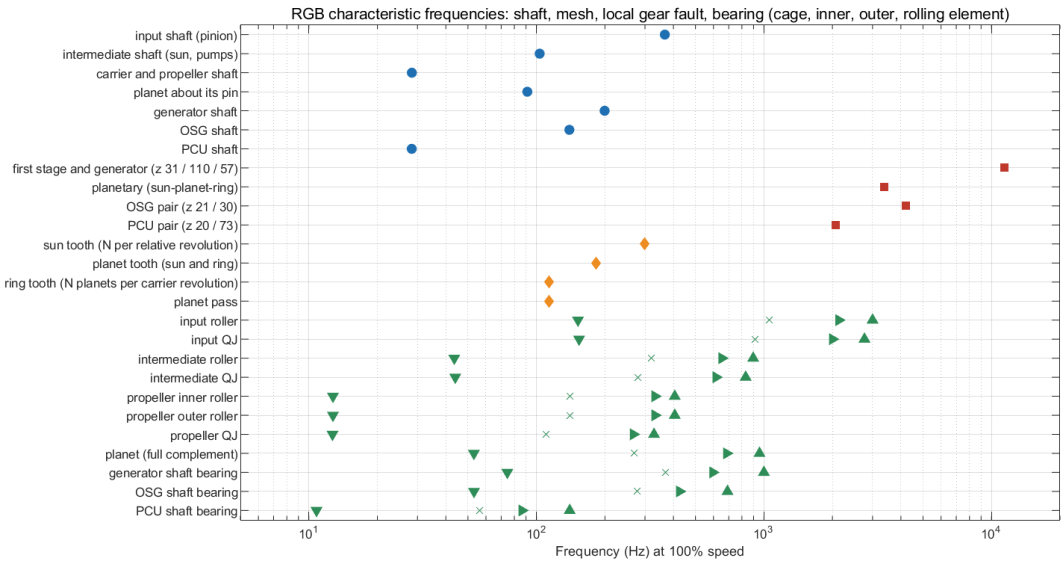


Fig. 54 Characteristic frequencies of the shafts, meshes, local gear faults and bearings at 100% speed (log scale).

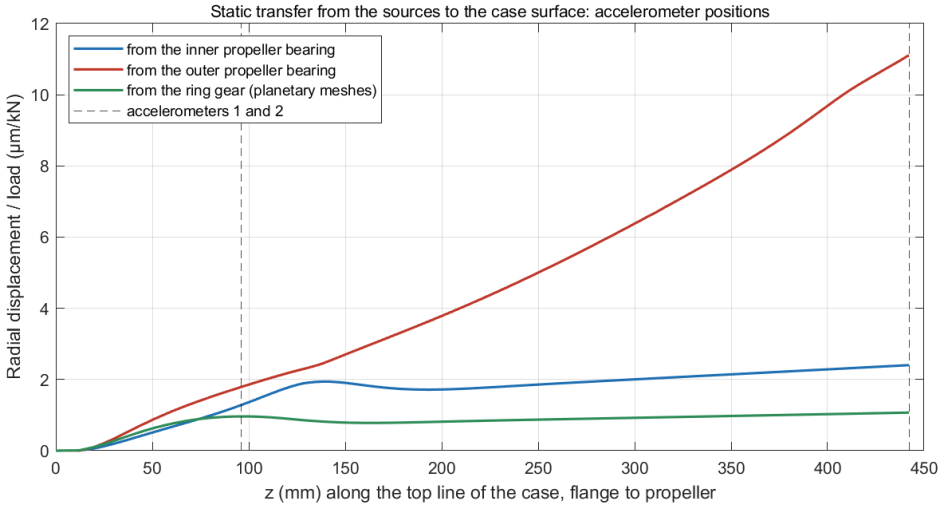


Fig. 55 Static transfer from the bearing seats and the ring-gear seat to the case surface along the case; the dashed lines mark the two accelerometer positions.

C. Indicators and Thresholds on a Synthetic Signal

To catch the faint signature of a cracked tooth in the loud, healthy noise of a gearbox, the signal is averaged in step with the shaft that carries the tooth. It is like photographing a spinning wheel with a strobe that flashes once per turn: the wheel seems to stand still, everything locked to it stands out, and everything else blurs away.

Without an engine, indicators were tested on a synthetic accelerometer signal built from the design kinematics: the mesh vibration of each planet, weighted by a window that peaks as that planet passes the sensor [45], the first-stage mesh, shaft orders and noise. Two faults were seeded: a sun tooth that loses stiffness when it meets each planet, and a spall on a planet bore that rings a case resonance. Four indicators were evaluated: FM4 and NA4 on the TSA of the sun relative to the carrier [46], a sideband energy ratio, and an envelope indicator at the planet-bore defect frequency of 685 Hz [47]. Fig. 56 shows what the two faults look like.

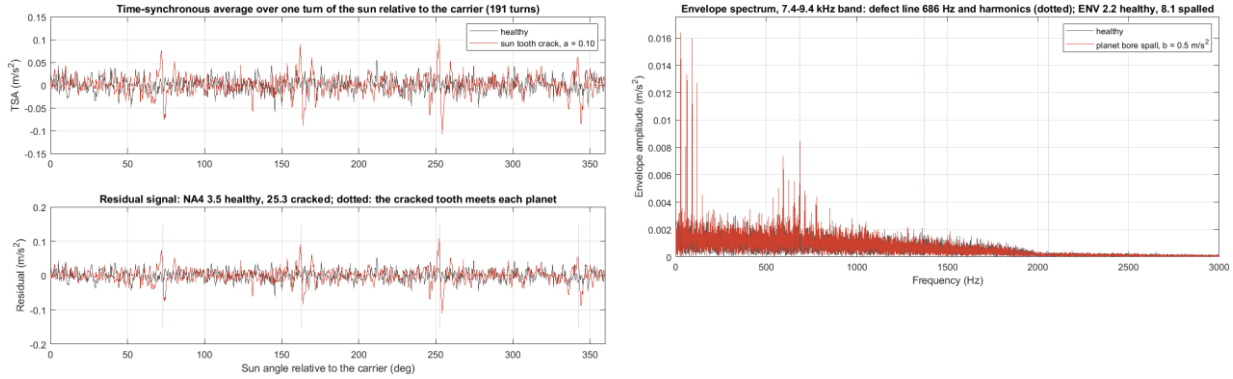


Fig. 56 Left: time-synchronous average and residual signal, healthy and with a cracked sun tooth; the crack shows each time the tooth meets a planet. Right: envelope spectrum, healthy and with a spalled planet bore; the defect line at 685 Hz appears.

Thresholds come from 50 healthy records, taking the logarithm of each indicator as normally distributed, at false-alarm probabilities of 10^{-3} (caution) and 10^{-6} (alarm) per record:

$$x_{thr} = \exp(\mu_{\ln x} + z\,\sigma_{\ln x}), \quad z = \sqrt{2}\,\mathrm{erfc}^{-1}(2\,P_{fa}) \tag{4}$$

FM4 and NA4 detect the crack above the alarm threshold at the smallest seeded size (5% amplitude loss), and the envelope indicator the spall at an impulse of 0.5 m/s² (Fig. 57). The sideband energy ratio is weak here for a known reason: with 45 sun teeth and four planets, the planet meshes cancel each other at the mesh frequency itself. The engine test will replace the synthetic baseline with at least 30 records per torque and speed bin.

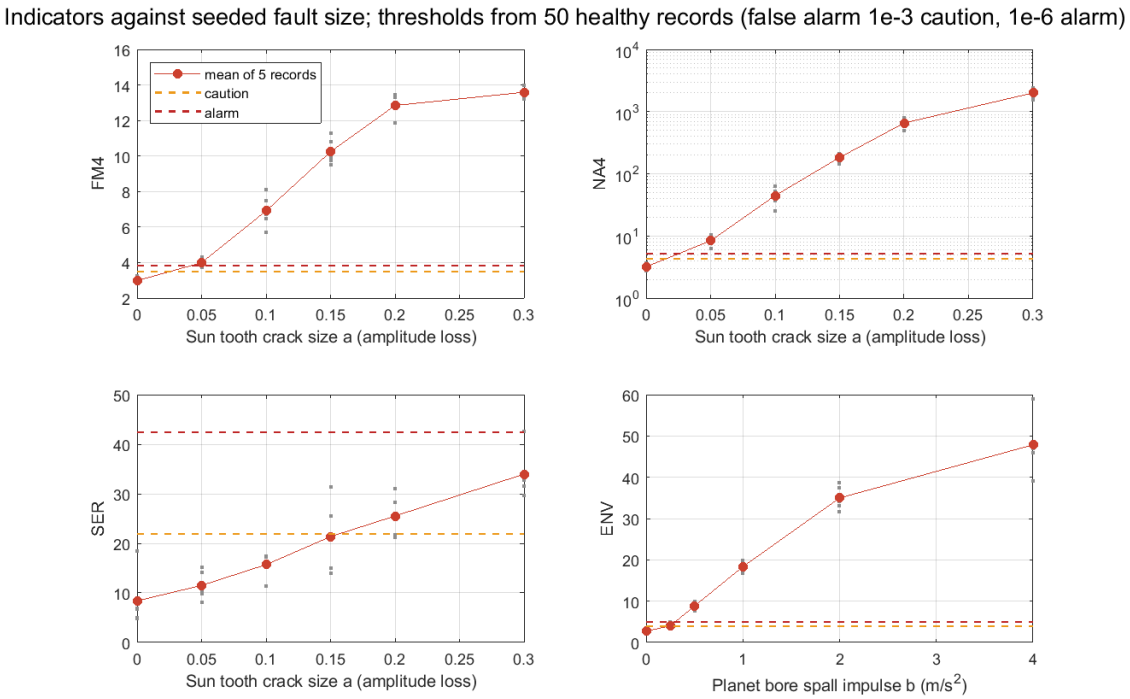


Fig. 57 Condition indicators against seeded fault size, with the caution and alarm thresholds from 50 healthy records.

X. Verification Plan

An analysis is a prediction; a test is the answer. At the CDR "met" meant "shown by analysis or inspection"; nothing had been tested, and the tests had no plan. The verification matrix now gives every requirement its methods, its evidence and, where it calls for a test, at least one of 25 planned tests; 39 requirements call for a test. The tests are ordered so that the cheap, early ones retire the largest risks before an engine is put at risk (Fig. 58); Table 13 lists the five test levels and what each of them closes.

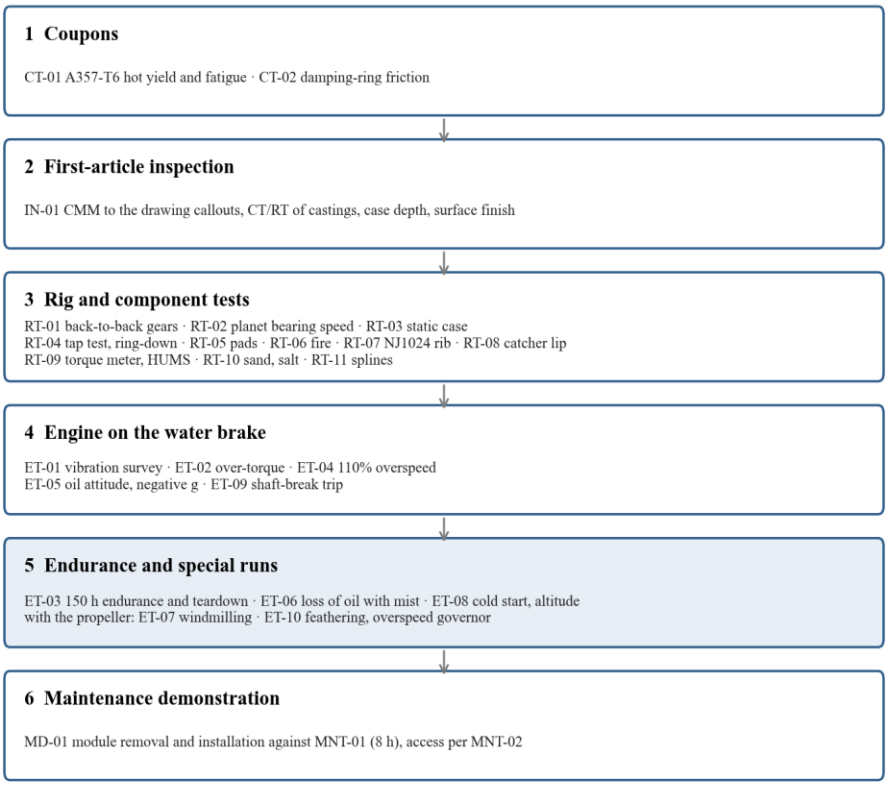


Fig. 58 Test sequence: 25 planned tests from coupons through rigs to the engine.

Table 13 The test levels and what each closes.

Level	Tests	What they close
Coupon	CT-01, CT-02	A357-T6 hot properties (casting factor 1.25), damping-ring friction ($\mu \geq 0.15$)
Inspection	IN-01	first article: drawing callouts, casting porosity, case depth, surface finish
Rig	RT-01 – RT-11	back-to-back gear rig (load sharing, transmission error, scuffing), planet bearing speed, static case test, tap test and ring-down, accessory pads, fire, standby bearing rib, catcher lip, torque meter and HUMS baselines, sand and salt, spline wear
Engine	ET-01 – ET-10	vibration survey (CS-E 650), over-torque (CS-E 820), 150 h endurance with teardown (CS-E 740), overspeed, oil attitude and negative g, loss of oil with the mist, cold start and altitude, shaft-break trip, windmilling, feathering
Demonstration	MD-01	module removal and installation against the 8 h target, access

The engine tests run first on a water-brake dynamometer in place of the propeller, which absorbs and measures the shaft power; windmilling and feathering need the real propeller. The test cell has nine groups of measuring points, from the torque meter checked against the dynamometer load cell to the three accelerometers sampled at 51.2 kHz and telemetry strain gauges on the gear web. Many values the design depends on can only come from partners; the interface control document lists 34 of them with their assumed values and the margin of the design to each (Fig. 59).

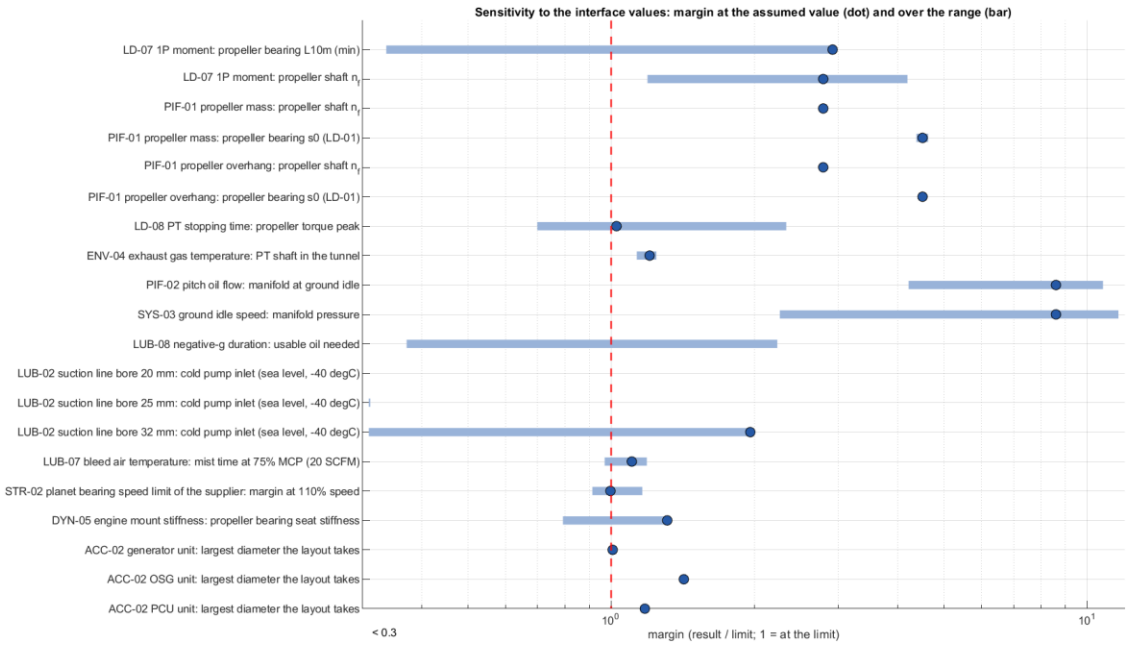


Fig. 59 Margin of the design to each interface value at the assumed value (dot) and over its plausible range (bar); 1 means at the limit.

XI. Comparison with Reference Engines

A design can meet all its own requirements and still be poor, if a flying engine does the same job with much less. Is 110 kg reasonable for a gearbox of this power? Neither type certificate data sheet gives the gearbox mass, so it was derived from the dry weights in the same sheets [3, 2]. The CT7-9B/C (365.1 kg) minus the CT7-2A turboshaft on the same core (194.6 kg) gives 170.5 kg, an upper bound because it also contains the drive shaft and the front-drive structure. The PT6A-67A is 22.7 kg heavier than the PT6A-66 for 66% more propeller torque; if all of that is gearbox and gearbox mass grows with torque to the power 0.75–1.0, the PT6A-67A gearbox weighs 57–72 kg.

Table 14 Comparison with the CT7-9 and the PT6A-67A (gearbox masses of the reference engines derived from data-sheet dry weights).

Quantity	This design	CT7-9B/C	PT6A-67A
Reduction ratio	12.93	15.90	17.58
Take-off power, kW	895	1,305 (1,394 APR)	895
Propeller torque, N·m	5,027	9,004	5,028
Gearbox mass, kg	109.95	170.5 (upper bound)	57–72 (estimate)
Power density, kW/kg	8.14	7.65	12.5–15.7
Torque density, N·m/kg	45.7	52.8	70–88
Layout	offset + planetary	integral, ahead of core	2-stage planetary
Efficiency at maximum continuous power	98.71%	not published	not published
Engine dry mass, kg	≈315	365.1	229.5

Table 14 compares the three gearboxes and Fig. 60 places them on one chart. This gearbox sits at the top of the band of the CT7-9 gearbox scaled to the same torque (95–110 kg) and has a slightly higher power density (8.14 against 7.65 kW/kg). It is 40–50 kg heavier than the PT6A-67A estimate, and the difference is exactly what the PT6A gearbox does not carry: the offset first stage (21.8 kg), the accessory drives (10.1 kg), the separate oil and mist hardware (4.7 kg) and allowances (7.0 kg). The planetary module alone, 66.4 kg with the propeller shaft, falls inside the PT6A-67A band. The installed engine would be about 315 kg dry, 85.5 kg more than the PT6A-67A, which at the same take-off

mass equals 3.2% of the 2,700 kg maximum payload of the Heron TP [1]. GE states no fixed overhaul interval for the CT7 family [48]; a sibling of the PT6A-67A has a 6,000 h initial interval [49].

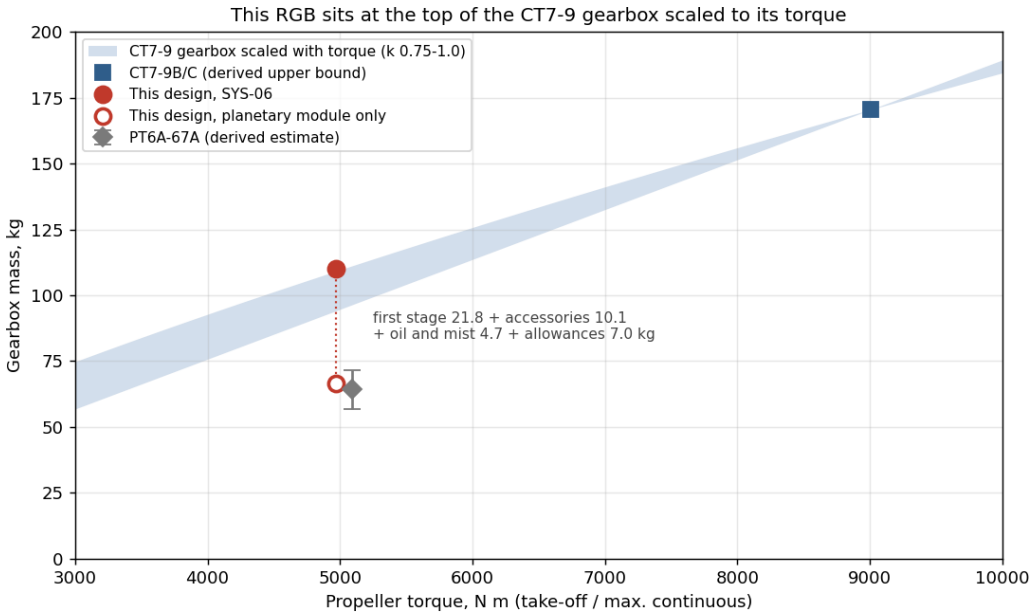


Fig. 60 Gearbox mass against propeller torque: this RGB, its planetary module, the CT7-9 and the PT6A-67A.

XII. Requirement Closure and Open Issues

The final version of the requirements folds in the Phase 3 changes: three new requirements (input-drive mass, loaded transmission error, thrust floor and limp mode) and ten changed (among them the 110 kg mass limit, the single oil grade, the run-dry definition, the protection layers and the 8 h module change). Of the 69 requirements, 34 are met by analysis, inspection or design, 19 partly, 7 are open and 9 wait for a value from the customer or an interface partner; none is at risk (Fig. 61). Fifty-two will be closed by a planned test or demonstration, 19 by analysis, inspection or design, and 26 wait for an interface value.

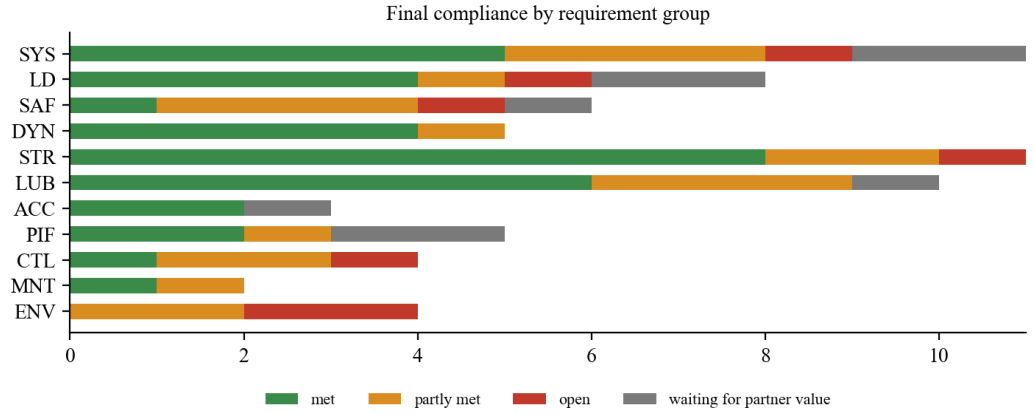


Fig. 61 Final compliance by requirement group.

A final review found no issue that stops the design, but four points limit how far its results can be trusted and must be read with them:

- **Mass margin.** The roll-up of 109.95 kg leaves 0.05 kg against the 110 kg limit, which is itself provisional. Several Phase 3 additions are booked by calculation rather than by weighing the CAD model.

- **Mount reaction.** Because the propeller turns opposite to the power turbine, the whole-gearbox reaction at the mounts is the sum of the propeller torque and the power-turbine torque, 7.45–7.49 kN·m at the transient torque, not the 6.4 kN·m entered in the interface table. The case flange bolts keep about 2.5 times the clamp force they need, so the effect is probably small, but the mount checks were not repeated.
- **Fire sealing.** The joint O-rings exceed their rating in the standard flame; the fire-rated secondary seal is proposed but unproven, and SAF-05 stays open until the fire test.
- **CAD currency.** The carrier plates, the flange bolt circle and fillets, the damping-ring groove and the nose parts of Phase 3 are in the calculations and drawing tables but not in the CAD model, so the drawing views and the visualisations show the Phase 2 geometry.

Many values the design rests on will be known only after the tests or from the partners. Table 15 states in advance what is done if one of them comes out worse than assumed, so that a bad result leads to a planned action rather than a surprise.

Table 15 What is done if a value comes out worse than assumed.

If…	then…	When
a measured or calculated torque spectrum gives a higher K_A	the full search is repeated; in Phase 1, K_A = 1.35 added 1.2 kg, more than today's 0.05 kg margin, so a mass decision follows	engine tests (RT-09, ET-02)
the first-stage flank modifications cannot be made	f_{ZCa} = 1.2 is used, about +2.4 kg, and again a mass decision follows	manufacturing planning
the aircraft wants the propeller to turn the other way (SYS-07)	two idler gears in the first stage; the offset changes, and the idler teeth, loaded on both flanks, are allowed about 30% less stress	aircraft data
the once-per-revolution propeller moment (LD-07) exceeds the assumed 2 kN·m	the propeller shaft and bearings are rechecked; the bearing life holds up to about 4,900 N·m (ICD-10)	propeller data
the exhaust heats the gearbox more than the estimated 3 kW	the cooler grows (ENV-04); in gas up to 800 °C the tunnel shaft stays at 132 °C against a 150 °C limit	core gas temperature (ICD)
the friction under the mist in the run-dry test is 1.5 times the lubricated value	LUB-07 is missed (3.5–3.6 min at 75% power); more mist air (about 1.7 m^3/min) or a stronger heat path in the case, both costing mass and core air; the author chooses	after ET-06
the gearbox exceeds 110 kg (margin 0.05 kg)	lightening is sought first; otherwise the limit rises, at most to 115 kg, by a decision of the author (SYS-06, TBR)	every design change
the fire test fails	wall-spray jets (about 0.05 kg), a fire shield over the first-stage case (about 0.4 kg), a fire-rated secondary seal and steel washers (about 0.17 kg); together more than the mass margin, so SYS-06 is reopened too	after RT-06
the loaded transmission error exceeds 0.15 µm (STR-11)	up to 0.167 µm STR-01 is still met; beyond that the profile modification is re-tuned, and failing that the face width or module grows and the planetary stage is re-laid out, at a cost in mass	after RT-01
the planet-bearing supplier cannot reach 110% speed (at least 6,020 rpm)	this is the narrowest interface (margin 0.997 with a similar catalogue bearing, ICD-25): an approved planet set is requested and its speed shown on rig RT-02; otherwise the planet bearing is redesigned	supplier offer

XIII. Conclusions

A complete, review-gated design of the power transmission of a pusher turboprop derived from a T700-class core has been carried from customer needs to a verification plan. The main conclusions are:

- Rear drive is the most suitable pusher installation for a straight-through, front-drive core: it keeps the gas path and the customer airframe layout at a cost of about 2–3% in fuel consumption.

- At the unusually low ratio of 12.94:1, an offset helical first stage followed by a single slow planetary stage avoids the fast planetary stage of the in-line layout. This permits rolling planet bearings and an emergency mist run-dry capability, and provides the accessory pads and the pitch-oil path naturally.
- The detailed design meets its strength, life, dynamic and efficiency requirements by analysis with stated margins: lowest pitting safety 1.28, root safety 1.55 with dynamic load, planet-bearing life 14,500 h, efficiency 98.71%, case Goodman factor 1.19, propeller-release cut sets at most 2.1×10^{-9} per flight hour.
- The design weighs 109.95 kg, at the top of the CT7-9 trend scaled to its torque; the mass beyond a PT6A-style gearbox is explained by the offset stage, the accessory drives and the separate oil system that the installation and the customer require.
- The remaining uncertainty lies in values that only tests and partners can provide: transmission error, damping-ring friction, case fatigue at the flange junction, fire sealing, and propeller and core interface data. The 25-test plan is arranged to retire these on coupons and rigs before an engine is put at risk.

The next steps are to update the CAD model to the analysed geometry, repeat the mount checks with the corrected reaction, reissue the interface table and close the mass margin before any further addition.

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Appendix A: Decision Log

Table 16 lists every decision recorded at the reviews. Most were taken by the author between options with their trade-offs; those marked "proposal" were recommendations that the author accepted at the review without change. The options and the reasoning behind each are recorded in the phase reports.

Table 16 Decisions recorded at the reviews (author: chosen by the author among options; proposal: recommendation accepted without change).

Phase	Topic	Decision	Basis
0	Customer aircraft	Heron TP-class MALE UAV that flies today on a PT6A-67A	author
0 (SRR)	Transient torque (LD-01)	137.5% for 20 s, as on the PT6A-67A (the alternative was 118% for 12 s, as on the CT7-9)	author
0 (SRR)	Over-torque test (LD-02)	CS-E 820 test at the LD-01 value	proposal
0 (SRR)	Gyroscopic rates (LD-05)	2.5 and 1.0 rad/s, the general values of 14 CFR 23.371	proposal
0 (SRR)	Reverse thrust (PIF-04)	not required; no beta mode	proposal
0 (SRR)	Run-dry (LUB-07)	30 min as a target, not mandatory (redefined in Phase 3)	proposal
0 (SRR)	Flat rating (ENV-01)	53 °C, as on the PT6A-67A	proposal
0 (SRR)	Life (STR-05)	TBO 4,000 h, service life 20,000 h	proposal
0 (SRR)	Electrical power (ACC-02)	the generator pad of the CT7-9, about 52 kW	proposal
1	Installation	C, rear drive (T700-701K style)	author
1	Gearbox architecture	O, offset helical first stage + planetary stage	author
1	Oil system	separate gearbox oil system, as on the CT7-9	author
1 (PDR)	Application factor	$K_A = 1.25$ until a measured or calculated torque spectrum exists	proposal
1 (PDR)	Shaft-break protection	detect within 20 ms, cut the fuel within a further 25 ms; turbine rotor to withstand 1.05 times the peak speed	proposal
1 (PDR)	Loss limit	at most 13.6 kW including the pumps, excluding the accessory power	proposal
1 (PDR)	Power and torque limit	895 kW without accessory load; torque limit on the power-turbine shaft	proposal

Phase	Topic	Decision	Basis
1 (PDR)	Rotation direction	propeller opposite to the power turbine; the other direction with two idler gears	proposal
1 (PDR)	Critical-speed rule	20% margin for forward whirl (DYN-01); a response limit for backward whirl (DYN-05)	proposal
1 (PDR)	Accessory pads	generator, overspeed governor, propeller control unit, oil pumps	proposal
1 (after PDR)	Planet count	four planets, 31/110 | 45/37/119	author
2	Propeller shaft	120/100 mm thin-walled shaft, no squeeze-film damper	author
2	Planetary margin (Decision 1)	superfinishing with the stiff planet bearing	author
2	Cold-climate oil (Decision 2)	MIL-PRF-7808 only up to 105 °C oil inlet (replaced in Phase 3)	author
2	Planet bearing (Decision 3)	70 mm face width, 8 mm rim, full-complement rollers	author
2	Propeller retention (Decision 4)	internal Ti-6Al-4V retention tube	author
2	Cooling air (Decision 5)	electric fan on the ground, ram air in flight	author
2	Case material (Decision 6)	A357-T6 sand casting	author
2	Innovation package (Decision 7)	maintenance-driven design and condition monitoring	author
2 (CDR)	Requirements v3	approved	author
3 (3.1)	Mass limit	raised from 100 to 110 kg (TBR) after lightening package A	author
3 (3.2)	Run-dry	at least 6 min at 75% power with an emergency air-oil mist	author
3 (3.3)	Oil grade	MIL-PRF-23699 only, ground preheat at cold bases	author
3 (3.3)	Input drive	separate mass limit of 6 kg (TBR)	author
3 (3.5)	Engine mounts	on the planetary-case flange; first-stage rear wall 16 mm	author
3 (3.5)	Second mass package	10 mm carrier plates, bottom-up fastener count	author
3	Requirements v4	approved	author

Appendix B: Repository and Tools

All tools, data and models are kept in the project repository: MATLAB calculation tools in calc/ (one folder per step) with expected test values in calc/tests/ produced by independent Python references in calc/reference/; the rotordynamic reference model (ROSS) in rotordynamics/; CATIA V5 scripts, parameter files, part and assembly models and drawings in cad/; requirement versions, compliance matrices, the interface control document and the verification plan in requirements/; materials, tolerances and the bill of materials in manufacturing/; safety analyses in safety/; maintenance and monitoring data in maintenance/; the reference-engine comparison in comparison/; and the figures, phase notes and this report in docs/. Repository: https://github.com/tanboga20/turboprop-from-turboshaft

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